

indicate that the Qumran community was enormously wealthy. Since this is an unacceptable concept for an ascetic society, either the Copper Scroll must be a product of a Qumranite crank, or it originates from elsewhere and was placed in a Qumran cave after the community ceased to exist. Moreover, Golb accuses the scholars controlling the publication of the scrolls of forcing their Qumran-Essene view on others. Allegro was ostracized because he did not follow to the 'official' line formulated by de Vaux in his colleagues.

Golb hypothesizes that during the siege on Jerusalem the manuscripts were brought to the Judean wilderness from various libraries and were hidden with the aid of the local inhabitants. They represent Jewish writings of three inter-testamental centuries.

Golb agrees with Dupont-Sommer that the Copper Scroll was formulated for bookkeeping purposes. According to Golb, the Scroll is a genuine document. It is the only autograph text found in Qumran which was copied, and the duplicate was hidden away for safekeeping. The Scroll refers to eight locations where documents or books were hidden, and these were probably similar to those found near the Dead Sea. The origin of the treasures, artifacts, and books referred to in the Scroll is Jerusalem. The treasures were presumably removed from the capitol before and during of the siege between 68-70 CE. For Golb the Copper Scroll is the most important discovery of Qumran; is the critical document disproving the Qumran-Essene hypothesis of the origin of the Dead Sea Scrolls.⁵⁵

Wise says that the Copper Scroll is crucial to identify who were the authors of the Dead Sea Scrolls. If the Scroll is linked with the Temple, then some, if not all, Dead Sea Scrolls may be connected to the Temple. This possibility could have influenced those who suggest a later date for the Scroll. The Scroll may list the hidden treasure of Herod's Temple, and it is likely that the Romans retrieved it in 70 CE.⁵⁶ Both Golb and Wise rely on Wilmot's unfinished dissertation. Wilmot considers the Copper Scroll as an authentic document, and seeks to establish its documentary and bookkeeping nature.⁵⁷

Garcia Martinez and van der Woude question Golb's hypothesis. They believe that the Copper Scroll is of Qumranite origin, and that it is part of the

55 "The Problem of the Origin and Identification of the Dead Sea Scrolls," *PAPS*, 124 (1980), 6b, 20 n. 50; "Who Hid the Dead Sea Scrolls," *BA*, 48 (1985), 68-82; "The Dead Sea Scrolls, A New Perspective," *The American Scholar*, 58 (1989), 177-207; *Who Wrote the Dead Sea Scrolls* (1995), 99, 117 ff, 151 ff, 273 ff, 327 ff, etc. See also, A. Dupont-Sommer, *The Essene Writings from Qumran* (1961), 383. B.W.W. Dombrowski, "Golb's Hypothesis: Analysis and Conclusions," *QCH* 5 (1995), 91-92.

56 "The Dead Sea Scrolls, Part 1 ...," *BA* 49 (1986), 140-54; "The Dead Sea Scrolls, Part 2 ...," *BA* 49 (1986), 228-42; "A List of Buried Treasure (The Copper Scroll)," *The Dead Sea Scrolls: A New Translation* (1996), 188-91.

57 *The Copper Scroll: An Economic Document of First Century Palestine*, prepared under the supervision of N. Golb at the University of Chicago, it will be published by M. O. Wise.

Qumran library. It is a real document as indicated by its dry realistic style, as well as the fact that it is recorded on copper.⁵⁸

Wolters vocalizes three words in the text as if they are in the first person. He suggests that "the composer of the *Copper Scroll* must have been a leading priest, perhaps the High Priest, of the Qumran community."⁵⁹

McCarter says that the treasure of the Scroll was probably real. He associates the wealth listed in the text with the Temple of Jerusalem and not with the Qumran community.⁶⁰

Wacholder discusses the prominent figure Zadok, who is mentioned in the Copper Scroll, as well as in the Temple Scroll and Damascus Document. He says that "the most cogent argument is that the unique monument described in 3Q15, 11:2-7 records a fictive memorial to the founder of Qumran ... although the uncertainties surrounding the comprehension of the Copper Scroll are sufficiently serious as to preclude citing this text as support for our identification of Zadok, the evidence for Zadok as the founder of the sect in other texts ... does lend validity to the hypothesis that the Copper Scroll was engraved by a sectarian craftsman." He adds that "CD 5:5, when coupled with the Copper Scroll and the rabbinic and Karaite traditions, makes it virtually certain that Zadok is a real person."⁶¹

Goranson's says that the Dead Sea Scrolls, including the Copper Scroll, belonged to the Qumran Essene library. The Scroll is considered to be a real document, and indicates that Essenes also lived east of the Jordan river.⁶²

Stegemann suggests that the Copper Scroll can be regarded as a register of those Temple bank holdings that survived the destruction of Jerusalem, and remained buried. Each entry with the Greek letter sets indicate the ownership of that deposit in the Temple bank. The first two Greek letter sets refer to Kenedaios and Chageiras, members of the royal house of the Adiabene family who converted to Judaism, supported the revolt against Rome, and are known from the works of Josephus. These were their bank deposits. Furthermore, there is no connection between the fourteen leather manuscripts found in Cave 3 and that of the Copper Scroll (3Q15). This cave was used by different people as a hiding place.⁶³

58 "A 'Groningen' Hypothesis of Qumran Origin and Early History," *RQ* 14 (1990), 526-36.

59 "Notes," 594; "History and the Copper Scroll"; "History," 292-93; Wolters, 36-37, 46-47, 52-53.

60 "Mysteryⁱ," 52-53; "Mysterious," 63-64a; "Mysteryⁱⁱ," 238-41; McCarter, 140-41.

61 *The Dawn of Qumran* (1983), 100, 153.

62 "Sectarianism, Geography, and the Copper Scroll," *JJS* 43 (1992), 282-87; "Further Reflections on the Copper Scroll," a paper submitted to the *International Symposium on the Copper Scroll* (1996).

63 *The Library of Qumran* (1998), 73-74; "The Qumran Essenes," *The Madrid Qumran Congress* 1 (1992), 98; see also McCarter, 147-48.

Mansoor mentions that among the possibilities is, that the copper sheets were originally designed for something else. When the Romans closed in on Jerusalem, someone engraved the present text to mislead them.⁶⁴

One can summarize the various hypotheses concerning the origin of the Copper Scroll as follows.⁶⁵

- (1) The Copper Scroll originates from the library of the sectarians of Qumran (whether they were Essenes or not). (a) It may describe the wealth of the sectarians. Or (b) it may be fictional.
- (2) The Copper Scroll originates from the Temple. (a) It was deposited in one of the Qumran caves which were used as repositories (*genizot*) controlled by the Jerusalem priesthood. Or (b) someone (a priest or a Levite) who had access to the Scroll brought it with him when he joined the Qumran sectarians. It was later stored among the sectarian documents in one of the caves of Qumran.
- (3) The Copper Scroll was composed to mislead the Romans when they closed in on Jerusalem.
- (4) The Copper Scroll is a register of the holdings of the Temple bank still lying buried after the destruction of the Jerusalem.
- (5) The Copper Scroll lists the accumulation of wealth soon after the destruction of the Temple hidden in anticipation of the rebuilding of the Temple.
- (6) The Copper Scroll belonged to Temple of Bar-Kokhba. It describes various holdings of that temple.
- (7) The Copper Scroll originates from the middle ages. It has no connection with the Temple or with the Essenes.

AUTHENTICITY OF THE COPPER SCROLL

The Copper Scroll is a dry list of over sixty distinct deposits of wealth hidden in various locations throughout Roman Palestine. Its genre is a typical list format. Wilmot demonstrates that the Copper Scroll resembles the Greek inscriptions of the Temple inventory document of Isle of Delos.⁶⁶ However, there are a wide number of historical documents which have list formats and bear similarity to the Copper Scroll. These include: Megillat Ta'anit (noted by Wilmot); the dedication of the Tabernacle (Num. Ch. 7); the list of the Canaanite kings destroyed by Joshua (Josh. Ch. 12); the list of the forbidden work on Sabbath (m. Shab. 7:2); the Bethphage ossuary lid (noted by Wilmot), Murabba'at 8 (noted by Wilmot); a large collection list among the Elephantine Documents (Cowley 22, 7 cols, 133 lines). Lists also occurs in Akkadian,⁶⁷ in Ugaritic,⁶⁸ and other sources.

⁶⁴ *The Dead Sea Scrolls* (1964), 50.

⁶⁵ Wolters presents a detailed discussion of the various suggestions of the scholars ("Apocalyptic," 147-54; "History," 285-92).

⁶⁶ "The Copper Scroll of Qumran (3Q15) and the Greco-Roman Temple Inventories."

⁶⁷ E.g., two economic texts (RS 1957.3; RS 1957.4; see M.C. Astour, "A Letter and

The last paragraph of the text provides critical information regarding the Copper Scroll. It refers to the existence and location of a detailed copy of the Scroll, which is only an abbreviated version (see #60, 12:11-13).

One can ask: Why were two copies of the same secret document? Why was it necessary to have an abbreviated and a detailed copy? Why was one engraved on copper sheets, and the other presumably written on parchment?

In response one can speculate that perhaps two groups controlled the treasures listed in the document, one having the Copper Scroll and the other its detailed copy. Since one copy might have complemented the other, both were needed to retrieve the hidden material. This could then have been a protective device to deny access to the treasures by unauthorized persons, such as Roman or local government officials, even Levites, or priests. Therefore, the Scroll should be viewed as being neither as a complete list nor a clear guide to locate the assets described in it. Rather it represents a brief coded summary of a more comprehensive documentary record of hidden wealth.

The Copper Scroll is probably the longest metallic inscription yet discovered. Using an expensive material for its preparation, and a laborious, time-consuming method to inscribe the text, it must have been an extremely important and valuable document. The reference to the hidden, detailed copy mentioned, significantly at the very end of the Scroll, demonstrates the importance of its text. Its cryptic content implies why a more detailed document was prepared, but unfortunately as yet has not been found.

THE HIDDEN TREASURE

According to the official announcement from Jordan, the Copper Scroll describes 200 tons of gold and silver buried in numerous places in Roman Palestine. This has long remained the widely accepted view. The text lists various objects, some are identified by numbers (e.g., vessels, gold bars, pots full of silver), some by weight (e.g., gold, silver), while others neither by numbers nor by weight (e.g., garments and/or covers, documents). In addition, there are several lacunae where numbers, numerals, or other impor-

Two Economic Texts," *The Claremont Rash Shamra Tablets* [1971], 31-35).

68 Since Ugaritic has close relation to Hebrew, one of these texts is cited, with its almost literal Hebrew equivalent, to demonstrate its style. *šb' kkr š'r b kkr addd wbkkr ugrt ḥmš kkrm alp ṭmn mat kbd d mnḥt* (= שבע ככר שעורים בככר אשדוד ובככר אגריט חמש בכרים אלף (= שמונה מאות כבד של מנהה in Hebrew), "Seven talents of barley for one talent of Ashdod; and for one talent of Ugarit five talents (barley). One thousand eight hundred heavy (jars) of cereals" (RS 1957.701 [obverse] 1-6); *ṭmnm nrt b ṭ kbd ksp šrt mqḥt w ṭlm kst šrm šin* (= אן שמונים נרות בשני כסף כבד עשרת מלקחים ושלושים כוסות עשרים צאן (= Eighty lamps for two heavy silver (shekel). Ten Tongs and thirty cups, twenty sheep" (RS 1957.701 [reverse] 1-5). This indicates that the talent of Ugaritic and talent of Ashdod were different weight, and that talent as a measurement were used for barley (M. Dahood, "An Economic Text," *The Claremont Rash Shamra Tablets* [1971], 31-35).

tant information were present, but are now missing. Therefore, it is impossible to calculate the actual *value* of the treasures listed in the Copper Scroll. The *amount* of the various metals listed by weight may, however, be calculated with some degree of certainty.

The scholars calculate the sum of metal listed by weight in the Scroll as follows. Milik: 2286 silver talents, 2040 silver and gold talents, 307 gold talents. Allegro and Lurie: 3282+ talents of silver (including mixed gold and silver, and unspecified metals), 1280 gold talents. Allegro (revised edition): 3179+ talents of silver (including mixed gold and silver, and unspecified metals), 385 gold talents. Pixner: circa 4500 talents of gold and silver.

The Copper Scroll does not contain any fables. The only issue cited by those who reject its authenticity is the enormous wealth listed in it. The suggestion that the Scroll lists 200 tons of gold and silver can be strongly challenged. The term “gold” occurs only eight times in the text, while “silver” twenty-one times. Some of these refer to various vessels not listed by weight. Thirty-six out of the sixty items do not mention the term gold or silver at all.

There are difficulties calculating the actual weight of a כֶּכָר, “talent.” Probably in the late Second Temple and in the Mishnaic era, the talent was equal to about 21.3 kg. The text uses both כֶּכָרִין and the otherwise unknown כֶּכָּ. The scholars agree that כֶּכָּ is an abbreviation for כֶּכָרִין, “talents.” Is this a valid assumption that can be substantiated? This is a fundamental question, since any conclusion as to whether the Scroll is a genuine document or not, depends on the interpretation of the amount of wealth listed in it. For an in-depth discussion, see Appendix A.

The Copper Scroll lists various silver, gold, and undefined types of objects or vessels. Many of them are כְּלֵי דְמַע, “*dema*’ vessels,” an otherwise unknown term. Some of these are identified by known Biblical Hebrew, Mishnaic Hebrew, and Judeo-Aramaic names, and were used in the Tabernacle and in the Temple. Therefore, כְּלֵי דְמַע most accurately can be rendered as “dedicated vessels” for the Temple, rather than “vessels containing heave offerings or priestly shares,” as suggested by many scholars. For an in-depth discussion, see Appendix D.

Many of the hiding places described in the Scroll were grave sites (e.g., נֶפֶשׁ, “tomb,” קֶבֶר, “grave”) and water-works (e.g., אֶמָא, “canal,” בֵּיב and בֵּיבָא, “drain”), while others could be either water-works or grave sites (e.g., מְעָרָה, “cave,” also means “ritual bath” and “underground grave site”; שִׁיחַ - שוּחַ, “pit, grave,” בּוֹר, “pit, cistern, grave”).

One may question: Why dedicated Temple vessels would be hidden in a ritually impure environment, such as a grave? A possible answer may stem from the fact that according to Tannaic sources, among the duties of the Court of Jerusalem prior to the three Pilgrim Festivals were the maintenance

of the highways, plazas, water-works (for drinking and purification purposes), and marking the grave sites (to avoid impurity; cf. m. Sheq. 1:1; t. Sheq. 1:2,4-5). It stands to reason that the complex water-works and grave sites controlled by the Jerusalem Court, would be considered particularly appropriate hiding places for Temple treasures, especially since metal vessels and garments could be purified.

WHY COPPER NOT PARCHMENT ?

A major question regarding this document is why it had to be engraved onto a one foot wide by eight feet long copper roll? The use of parchment would have been far more appropriate. A parchment document is much easier and economical to produce, it can be easily rolled up, is lightweight, portable, and can be readily hidden. Hammering letters into expensive copper sheets is a time consuming, noisy, costly, and difficult to correct or update. A scribe was more readily available to write out the text than an engraver. With frequent usage, a large copper roll with indented text, could crack and break into pieces. In addition, parchment does not need relatively large sized characters, as does copper sheet. Therefore, it is unlikely that the originators of this document used copper merely to assure that it would have an extended life span. They must have had more cogent reasons for doing so.

One may hypothesize that the Scroll might have been prepared specifically for the priestly leaders of Jerusalem. Scripture compels the priests, especially the high priest, to maintain a sanctified lifestyle and to avoid ritual contamination, such as contact with a dead body, grave site, impure person or object. The complex purification process included certain rites, such as the sprinkling of water containing the ashes from a red heifer and immersion of the impure person or object in a ritual bath.

According to Scriptures, ritual impurity can contaminate leather utensils.⁶⁹ The assumption is that skin documents are included in this category. On the other hand, based on Rabbinic sources, it is difficult to decide whether a document can become defiled, although Scriptures cannot become impure.⁷⁰ *שְׁטֵיט*), but according to the decree of the sages (*מִדְּרַבָּנָן*), they can become impure. See also Maimonides, *Tosephot Yom-Tov*, and *Hilkheta' Gevira'* on this Mishnah. *Mele'khet Šelomoh* says that if the leather is soft one can wrap it around a small item, it can become impure. He refers to *Tosaphot*, which

69 Cf. "and as to every garment, and everything made of skin, and every work of goats hair, and everything made of wood, you shall purify" (Num. 31:20); and "and everything that does not endure fire you shall make pass through water" (Num. 31:23).

70 The Mishnah says, "כְּלִי עֵץ וְכֵל עוֹר וְכֵל עֶצֶם וְכֵל זָכָת" (wood utensils, and leather utensils, and bone utensils, and glass utensils), (when) they are flat, they are pure (i.e., cannot become impure), but (when) they are (formed as) receptacles (they can become) impure" (m. Kel. 2:1). Bertinoro points out that this Mishnah refers to the law of the Torah (*מִדְּרַבָּנָן*)

says that a soft leather piece can be used to wrap around a jewel or needle, thus it can become impure (b. Shab. 63b, Minnayin La'arig). See also *Rosh* on m. Kel. 27:1. Maimonides rules (*Mišneh Torah*, Hilkhot Kelim, 1, 10) that according to the decree of the Rabbis, flat leather utensils can become impure. Furthermore, this author discussed with several distinguished Rabbinical scholars as to whether a document written on leather is considered as a "flat leather utensil" or not; whether a document written on parchment, such as a letter of divorce (בטא), can become impure or not. A definite decision could not be reached.

Yet it is conceivable that the originators of the Copper Scroll considered ritually contaminated documents impure. Perhaps they were Sadducees who were known for their strict interpretation of the Bible, especially concerning ritual purity.⁷¹ According to the Talmud, many high priests and priests of the Second Temple era were Sadducees. Naturally, a parchment-like document would have been damaged, if not destroyed, by the purification rites, while an engraved copper document could readily be rendered pure.

Many hiding places referred to in the Scroll were burial sites. Therefore, not only the objects and the recovery team,⁷² but any documents placed there would also become defiled. As a result, the rolls would have had to be cleansed according to the strict purification laws before being returned to the priestly leaders in Jerusalem. If these assumptions are correct, a document engraved on copper could easily have been purified without damaging its valuable inscription.⁷³ Therefore, the use of copper sheets instead of parchment might have solved the problem of ritual impurity and permitted relatively easy purification, something not feasible for parchment.⁷⁴

71 This author, in July 1987, while working on his doctoral dissertation at New York University, submitted to *BAR* a lengthy article (as well as a disk), entitled "The Mysterious Copper Scroll — Is There a Solution?" Unfortunately the article (without the disk) was returned to his mentor, Prof. Schiffman, in January 1988, who passed it on to the author at the end of May 1988. *BAR* suggested revision of the article, as well as drastically shortening it. The author in November 1988 declined to do so, since he felt it would seriously diminish its value. In this article the author already suggested that perhaps the Copper Scroll was written by Sadducees.

72 Since Scripture prohibits Kohanim ("priests," pl. of Kohen) to enter grave sites (cf. Lev. Ch. 21; Num. Ch. 19; b. 'A.Z. 13a), priests could not participate in such teams, only Levites or other trusted persons.

73 The Mishnah says, "כְּלֵי מַתְכוֹת וּמַתְכוֹת" (metal utensils), their flat parts and their receptacles (may become) unclean" (m. Kel. 11:1). This may imply that a flat metal sheet having inscription on it, such as the Copper Scroll, could be contaminated.

74 Allegro suggests that the text may have been engraved on copper sheets to ensure its preservation, and it could have a ritual significance, "since it was declared that articles made of sheet metal are not susceptible to ritual uncleanness" (m. Kel. 11:3). He adds, "that would mean that if our scroll became in any way ritually defiled during its interment in a cave, it could, on rediscovery, be washed clean and fit for priestly handling" (Allegro, 63; Allegro^{rev}, 47-48). Apparently, there is a contradiction in these explanations.

The complex ritual impurities and purification processes are biblical commandments that are further expounded upon in the Talmud. טומאת ידים, "impurity of the hands," however, is a decree of the early Rabbis. This is a protective impurity, meant to prevent consecrated things from contamination outside the Temple. This statue serves, among others, to protect the priestly share from rats (b. Shab. 13b-14a; b. Pes. 19a). For instance, according to the Bible, a ritually clean priest may consume his priestly share anywhere. According to Rabbinical ordinance, if a ritually clean priest touches properly written Scriptures outside the Temple, his hands become "impure," and he is obligated to wash his hands before consuming his תְּרוּמָה, priestly share. Touching it without washing the hands renders the priestly share unfit for consumption. This enactment does not imply that the biblical books are impure. On the contrary, only certain holy objects, such as Scriptures, render the hands impure. Prayers that may contain small biblical segments, and non-canonical writings, such as the Book of Ben-Sira, do not make the hands unclean (t. Yad. 2:12-13; t. Kel. B.M. 5:8).

Rabbi Johanan ben Zakkai was ridiculed by the Sadducees on the subject of "impurity of hands." He replied them that the reverence for Scriptures creates their impurity (m. Yad. 4:6; t. Yad. 2:9). This is only one explanation for the enactment. Although the Sadducees opposed the Rabbinical decree of "impurity of the hand," they strictly adhered to the complicated biblical laws of purification. The Sadducees even extended the power of contamination to "indirect contact" (m. Yad. 4:7).⁷⁵ Furthermore, the Qumran sectarians interpreted the biblical commandments of ritual impurities and purification processes stricter than the Pharisees or Tannaim.⁷⁶

One may speculate that two bodies controlled access to the treasures listed in the Copper Scroll: the high priest having the Copper Scroll itself, and a group of leaders having its detailed copy; and both documents were needed to recover the hidden objects. But who were the members of this group? The various Talmud references suggest that an executive committee with a chain of command held control over the Temple treasury (such as, funds for maintenance, public sacrifices, cult vessels, public works) was: the king, the high priest, two controllers, seven trustees, and three treasurers (m. Sheq. 5:2; t. Sheq. 2:15; y. Sheq. 5:2).⁷⁷ One might reasonably assume that the treasures

⁷⁵ See also J.M. Baumgarten, "The Pharisaic-Sadducean Controversies about Purity and the Qumran Texts," *JJS* 31 (1980), 157-70.

⁷⁶ Cf. L.H. Schiffman, "The Impurity of the Dead in the Temple Scroll," *Archaeology and History in the Dead Sea Scrolls, The New York University Conference in Memory of Yigael Yadin* (1990), 135-56; J. Milgrom, "The Scriptural Foundations and Deviations in the Laws of Purity of the Temple Scroll," *ibid.*, 83-99.

⁷⁷ According to Maimonides (on m. Sheq. 5:2), the chain of command was: high priest, chief of the priests, two controllers, seven trustees and three treasurers.

listed in the Scroll were controlled in a comparable manner.⁷⁸

The last king of Judea was Agrippa I (r. 41-44 CE). After his death the Romans placed his brother, Herod II, king of Chalcis in the Lebanon (r. 41-48 CE), in charge of the Temple administration and treasury, and the right to appoint the high priest. In 50 CE Herod II's kingdom, as well as the right to appoint the high priest were transferred to (Herod) Agrippa II (= Marcus Julius, 28-92 CE), son of Agrippa I, but was not appointed as king of Judea. At the time of the Jewish revolt, Agrippa II fought on the side of the Romans.⁷⁹ Probably during this turbulent era the Scroll was written.

ḤAREVAH/ḤARUVAH AND THE VALLEY OF ACHOR

The first phrase of the Scroll is בְּחִרְוֵהָאִבְחִרְיָהָא שְׁבַעֲמֶמֶק עָכוֹר, "In/at/by (a/the) ruin/Ḥaruvah/Ḥarevah which is in/at/by (the) Valley of Achor-/Disturbance." It marks the beginning of the first description of sixty or so deposits listed at various places in Roman Palestine. There is good reason to assume that the Copper Scroll is a genuine document of actual wealth. Thus, Ḥarevah/Ḥaruvah or a/the ruin in the Valley of Achor must refer to a location at the time of the compilation of the document. This is especially true since the Valley of Achor is mentioned again in the text, rather than a nick name of a certain place, although one cannot exclude this possibility.

One may ask why the compiler(s) chose to start the text with a deposit located in Ḥarevah/Ḥaruvah or a/the ruin in the Valley of Achor/Disturbance? Why was not another place chosen to be mentioned at the outset of the Scroll? Was this done purposely, and if so, is it significant?

The very first word, חִרְיָהָ - חִרְוֵהָ, "ruin," connotes destruction of the Temple of Jerusalem, as Haggai prophetized, וְהַבֵּית הַזֶּה חָרָב, "while this Temple is in ruins" (Haggai 1:4), and as Nehemiah told to the Persian King Artaxerxes, הָעִיר בֵּית קְבָרוֹת אֲבֹתַי חָרְבָה, "the City of my forefathers' graves is in ruin" (Neh. 2:3). Likewise the Talmud uses the term חוֹרְבָן (MH), "destruction," e.g., חוֹרְבָן בֵּית ראשון וְחוֹרְבָן בֵּית שֵׁנִי, "the destruction of the First Temple and the destruction of the Second Temple" (b. Git. 57b).⁸⁰ Moreover, the third line of the Scroll starts with the difficult word אַרְיָה. The terms חִרְוֵהָ\חִרְיָהָ and אַרְיָה bring to mind verses of the תּוֹכַחְתוֹת (pl. of

78 Based on Rabbinic sources, it seems that many of the High Priests of the last era of the Temple were Sadducees, while the Chief of the Priests, a position equal to Deputy High Priest, was a Pharisee. Perhaps this arrangement might have been the result of an agreement between the two major Jewish factions. The Tosephta condemns the family ties of the trustees and treasurers of the Temple with several Sadducee high priests, and criticizes the anti-social acts of their servants (t. Men. 13:21). See also, #60, 12:11-13.

79 E. Elazary, "Agrippa I," *EJ* 2, 415-17; "Herod II," *EJ* 8, 387-88; A. Schalit, "Agrippa II," *EJ* 2, 417-18; S. Applebaum, "Herod I," *EJ* 8, 375-85, esp. 378, 385.

80 Cf. וּבְקֵץ חֲרָבָן הָאָרֶץ, "and at the end of the destruction of the Land" (CD 5:20).

(תוכחה), “reproofs,” pronounced by God if the Israelites will not adhere to His commandments.⁸¹

וְנָתַתִּי אֶת עָרֵיכֶם חֲרָבָה וְהַשְׁמֹתִי אֶת מִקְדָּשֵׁיכֶם וְלֹא אֶרִיחַ נִיחָחְכֶם: וְהַשְׁמַתִּי אֶת הָאָרֶץ ... וְהָיְתָה אֶרְצְכֶם שְׁמָמָה וְעָרֵיכֶם יִהְיוּ חֲרָבָה:

And I will put your cities into ruin, and I will make your sanctuaries desolate, and I will not smell your pleasing sacrificial odors. And I will devastate the land ... and your land shall be a desolation, and your cities shall be ruin (Lev. 26:31-33).

Furthermore, עֶמֶק עָכֹר, “Valley of Achor” or “Valley of Disturbance,” which is later in the Scroll called עֶמֶק עָכוֹן, “Valley of Achon,” brings to mind the sad story of Achan who committed a cardinal sin by stealing חֲרָם, “banned” articles from the booty during the onset of the conquest of Canaan by Joshua. Achan confessed his sin.

וַיַּעַן עָכָן אֶת יְהוֹשֻׁעַ וַיֹּאמֶר אָמְנָה אֲנֹכִי חָטָאתִי לַיהוָה אֱלֹדֵי יִשְׂרָאֵל וְכֹזֶאת וְכֹזֶאת עָשִׂיתִי: וְאֶרָאָה (כ', וְאֶרָא ק') בְּשַׁלָּל אֲדָרַת שְׁנָעַר אַחַת טוֹבָה וּמֵאֲתִים שְׁקָלִים כֶּסֶף וְלִשֹׁן זָהָב אֶחָד חֲמִשִּׁים שְׁקָלִים מִשְׁקָלוֹ וְאַחַד מֵדָם וְאַקָּחָם וְהֵנָּה טְמוּנִים בָּאָרֶץ בְּתוֹךְ הָאֶהָלִי וְהַכֶּסֶף תַּחְתֶּיהָ: וַיִּשְׁלַח יְהוֹשֻׁעַ מַלְאָכִים וַיִּרְצֻוּ הָאֶהָלָה וְהֵנָּה טְמוּנָה בָּאֶהָלוֹ וְהַכֶּסֶף תַּחְתֶּיהָ:

And Achan answered Joshua and said, “Indeed, I have sinned against the Lord, the God of Israel, and thus and thus have I done. And I saw among the spoil a goodly Shinar (or Babylonian) mantle, and two hundred shekels of silver, and one wedge of gold, fifty shekels its weight. And I desired them and took them. And behold, they are hidden in the earth in the midst of my tent, with the silver under it. So Joshua sent messengers, and they ran to the tent; and behold, it is hidden in his tent with the silver underneath it” (Josh. 7:20-22).

As a result, Achan and his family were stoned in a valley which was named after the incident as Valley of Achor or Disturbance (see #1, 1:1).

In addition, there are several parallels between the confession of Achan and the Scroll. מִשְׁקָל, “weight,” appears twice in the Scroll (see #1, 1:4); it occurs in the confession of Achan. Achan hid the stolen חֲרָם, “banned, consecrated,” articles consisting of an expensive mantle, silver, and gold in the ground. Similarly, the Scroll lists hidden precious metals, dedicated vessels and garments, and some of them are חֲרָם, “banned, consecrated,” objects (see #41, 9:10). The Scroll uses טְמוּנָה and חֲפֹר, synonyms of טְמוּנָה, “hidden” (sg. of טְמוּנִים; see #3, 1:7; #17, 4:7). Also, the word וְאֶרָאָה has an extra *he*, as the Masorah notes, instead of וְאֶרָא, “when I saw.” This may resemble the

81 There are two set of תּוֹכְחוֹת, “reproofs” (Lev. 26:14:45; Deut. 28:15:68); they are also called קְלָלוֹת (pl. of קְלָלָה, fem.), “curses” (m. Meg. 3:6; y. Meg. 3:7 [25b]; b. Meg. 31b). See “תּוֹכְחָה” in *OY* 10, 231b; A. Rothkoff, “Tokheḥah,” *EJ*, 15, 1192.

possible spellings of חריבא or חרובאה instead of חריבא, חריבא, חרובא, or חרובא (with an extra *he*).

Can all of these similarities be merely coincidental? Mentioning “ruin” and “Valley of Disturbance” may sound like a curse which could serve to scare away unwanted intruders. Can these resemble ancient Jewish, Phoenician, etc., funeral inscriptions where potential intruders are cursed?⁸²

On the other hand, Hosea and Isaiah prophetized that at future time to come (i.e., Messianic era), the Valley of Achor will cease to be a desert:

And I will give her (i.e., to Israel) her vineyard from there (i.e., from the desert) וְאֶת עֵמֶק עָכֹר (and [make] the Valley of Achor) into a Gate of Hope (Hos. 2:17).

And the Sharon will become a pasture for sheep, וְעֵמֶק עָכֹר (and the Valley of Achor) for cattle to lie down, for My people who have sought me out (Isa. 65:10).

Also, according to the Jewish tradition, the Messiah will be an offspring of King David. Curiously, some names in the Scroll can be associated with the family or immediate circle of King David. E.g., אֲשֶׁת־שְׁלֹמֹה, “Pool of Solomon” (5:6), and חֲרִיץ־חֲרוּצַי שֶׁל שְׁלֹמֹה, “Trench of Solomon” (5:8-9), vs. Solomon son of David (e.g., 2 Sam. 12:24); יֵד אֲבִשְׁלֹמֹה, “Monument of Absalom” (see #48, 10:12), vs. Absalom son of David (e.g., 2 Sam. 3:3); בֵּית תָּמָר, Beth Tamar (9:14-15), vs. Tamar daughter of David (e.g., 1 Chron. 3:9), and Tamar grand-daughter of David (2 Sam. 14:27); קֶבֶר צָדוֹק, “Grave of Zadok” (11:3), and גִּן צָדוֹק, “Garden of Zadok” (11:6), vs. Zadok the priest of David (e.g., 2 Sam. 16:27); בֵּית הַקֹּז, “House of Hakkoz” (7:9, alt.), vs. בְּנֵי הַקֹּז, “sons of Hakkoz,” one of the 24 priestly watches established by King David (cf. 1 Chron. 24:10).

Can the first three words of the Scroll, the contemporary ruin in the Valley of Achor, imply hope for the future? Can it allude to the consoling prophecies of Hosea and Isaiah? Can the first phrase of this document mean that better days will come when the hidden treasures will be retrieved and used again for the Temple? Can the biblical names Solomon, Absalom, Tamar, Zadok, Hakkoz, associated with the coming of Messiah? Obviously, these are merely attractive speculations, but they fit well into the historical time-frame when the document was written, and may add to its authenticity of the Scroll as a valid historic relic.

82 E.g., זאת [קברת ...] יהו אשר על הבית אין פה כסף וזהב [כי] אם [עצמותיו] ועצמות אמתה, אתה ארור האדם אשר יפתח את זאת, “This is [the tomb of ...]yahu, the manager of the palace. There is no silver or gold here, [but] only [his bones] and the bones of his maid are with him. Cursed shall be that man who opens this (grave site)” (see E.Y. Kutscher, “כתובות” in *Hebrew and Aramaic Studies* [1977], 170-171). Also cf. the Phoenician Ashmunezer funeral inscription (*KAI* 1, 3 [#14]).

SUMMARY

The Copper Scroll (3Q15) is an atypical historical document. It is primarily a detailed dry list of over sixty distinct deposits of wealth hidden in various locations throughout Roman Palestine.

The very last paragraph of the text provides a vital clue regarding the status of the Scroll itself. It refers to the existence and location of "a copy of this document with its explanation, with their measurements and specification for each and ev[ery] (item)." This clearly indicates that the *detailed copy* contained much more information than the Copper Scroll itself, which is only an abstract. Whether the detailed copy was also the *master copy* from which the Copper Scroll was copied or dictated, cannot be determined. In any case, essential data had intentionally been omitted from the Copper Scroll, probably to maintain the security of the treasures (see #60, 12:11-13).

During the later part of the Second Temple era frequently the foreign rulers appointed the high priest, and often the position was bought or secured by bribes by well-to-do rich priests who had good connections with the authorities.⁸³ Often high priests were not trusted by the Jerusalem Court.⁸⁴ If the Copper Scroll was indeed engraved for the high priest, this could be the reason why detailed information was omitted. Although the Scroll indicates the location of a detailed copy, it was hidden in a ritually impure grave site, which automatically prohibits its entry by priests. Thus, priests would be unable to recover it.

Careful study of the Copper Scroll leads one to confidently accept the authenticity of the Scroll's contents. There is nothing whatsoever in the document to suggest that its text may be fiction, except the apparent large extent of the wealth listed in it. The document refers to various types of dedicated vessels used for Temple services, many of them known from the Masoretic Text, Temple Scroll, Talmudim, Midrashim, etc. The text also mentions second tithe, seventh-year treasury, consecrated objects, and such names as Grave of Zadok, Garden of Zadok, and the Ford of the High Priest. All these references can be associated with the priesthood, and support the view that the Scroll is an authentic valid historical document.

The total wealth described in the Copper Scroll, has been shown to be much less than the inflated figure initially suggested by some scholars, al-

83 Cf. J.D. Eisenstein, "כֹּהֵן (כֹּהֲנִים; כֹּהֵנָה)" in *OY* 5, 258a-59a; I. Gafni, "High Priest (In the Second Temple Period)," *EJ* 8, 473-74; M. Stern, "Priests and Priesthood (From the Beginning of the Hellenistic Era Until the Destruction of the Temple)," *EJ* 13, 1086-88.

84 In similar situation the Elders of the Court swore the presiding High Priest strictly follow their orders regarding the Yom Kippur Temple services. The Mishnah dramatically describes the act of swearing which concluded by weeping of the High Priest, as well as Court representatives (m. Yoma 1:5).

though it is still a sizable amount. It is doubtful whether the so called sectarians of Qumran or any other secluded, ascetic community could and would have accumulated such assets. On the other hand, the Temple of Jerusalem was renowned for its wealth. Therefore, it is most probable that the document lists the hidden treasure of the Jerusalem Temple.

Many hiding places described in the Scroll were grave sites and water-works (see #26, 6:7). According to the Talmud, among the duties of the Court of Jerusalem in the Temple era prior the three Pilgrim Festivals were: maintenance of the highways, plazas, water-works (for drinking and for purification purposes), marking the grave sites (to avoid transmitting ritual impurity; m. Sheq. 1:1; t. Sheq. 1:2, 1:4-5). Undoubtedly, the complex water-works as well as the grave sites, having sanctity and controlled by the Jerusalem Court, were attractive hiding places for Temple treasures. Interestingly, several technical terms used in the Copper Scroll have more than one meaning, among others "grave site" and "water-work," and as such can be found in Rabbinic literature. All of this suggests Temple ownership of the listed treasures.

The engraving of the Copper Scroll, based on palaeographical evidence, is dated between the years 25-75 CE, i.e., the last few decades of the Temple. Since this was a turbulent period, one can assume that the trustees of the Temple of Jerusalem would seek to hide relatively small valuable objects and eventually formulate lists of the hidden treasures, i.e., the Copper Scroll and its detailed copy hidden elsewhere, as mentioned at the very end of the text.

This study offers a wide range of possible readings, vocalizations, and interpretations of the Scroll's text. Many of these are new and original, and are discussed in detail in the Commentary and Appendices. The author's preferred readings and its rationale are given. These are based on extensive usage of the Rabbinic literature which provides the most reliable source to secure a valid interpretation of the Copper Scroll. As a consequence of this approach, new light has been shed on the reading, vocalization, and interpretation of this valuable historic record. In addition, the possible reasons for its unique character, being inscribed on copper sheets, is now much better understood.

It is uncertain whether the detailed copy mentioned at the end of the text will ever be discovered. Unfortunately, without this copy the mystery of the Copper Scroll will never be fully solved. Nevertheless, this study provides new insights into and suggests possible solutions for a number of critical issues and vital questions, although many of them remain largely speculative.

This in-depth study also contributes to the knowledge of the Hebrew language used at the time of the destruction of the Temple of Jerusalem in general, and to the study of the Dead Sea Scrolls in particular. It may also help

to provide a better understanding of certain segments of the Masoretic Text, Talmudic and Midrashic literature. Hopefully, this work will stimulate others to adapt the approach of intensive utilization the extensive legacy of Rabbinic literature for the investigation of other Dead Sea Scrolls and Jewish historical documents as to better understand their meaning and significance.

CONCLUSION

The Copper Scroll lists various types of dedicated Temple vessels, many of them known from the Masoretic Text, Temple Scroll, Talmud, Midrash, etc. It also mentions second tithe, Sabbatical year produce, consecrated objects, as well as such names as Grave of Zadok, Garden of Zadok (may be associated with the high priest's families), and the Ford of the High Priest.

Much of the information, as well as many of the concepts and ideas discussed in this work, clearly imply that the Copper Scroll is a genuine document. It is most likely originated from the Temple of Jerusalem and describes hidden vessels, garments, and wealth of the Temple Treasury.

The Copper Scroll must have been engraved shortly before the destruction of the Jerusalem Temple. It was not done in haste in the last phase of the insurrection against the Romans. Secretly hiding the tremendous wealth listed in the Scroll, obviously took a considerable amount of time. In addition, it must have been a very tedious task to engrave twelve columns of text on three copper sheets, eight feet in total length.

The Copper Scroll is one of the most unique texts among the ancient documents in general, and the Dead Sea Scrolls in particular. Although its mysterious nature may never be uncovered, this extensive study in the context of relevant traditional Hebrew and Judeo-Aramaic, as well as other sources, sheds very substantial new light towards a better understanding of the Copper Scroll in its historical context.

APPENDIX A

THE ABBREVIATION ככ VERSUS ככרין

INTRODUCTION

The Copper Scroll consists of 60 items, and lists various objects, such as dedicated vessels, made of silver, gold, or unspecified metal (see Appendix D), precious metals (silver, gold, or unspecified), garments, and documents. These were hidden at various locations in Roman Palestine. The Scroll identifies the weight of the metals, as well as some of the vessels. Units of weight are mentioned almost 50 times, and are mainly found in the last phrase of an item.

The Scroll records about 2600 ככרין and 1060 ככ of metals. As suggested by Allegro¹ and Milik,² it is accepted that ככ is an abbreviation for ככרין, "talents."³ Accordingly, the Scroll list about 3660 talents of silver, gold, and unspecified material. The rest of the specified weights are relatively small, except the unknown value of 165 units of עֲשָׂתוֹת זָהָב, "unused gold masses."⁴

Regarding content, students of the Copper Scroll can be placed into two groups, those who consider it a work of fiction, and those who consider it a genuine document. Milik claims that such enormous wealth must be imaginary. Allegro resolves this problem by equating the talent with mina, thus reducing it to less than 2% of its actual weight, which in turn devalues the treasure by more than 98%. Lurie accomplishes the same goal by equating the talent with the Roman pound. Both of these simple solutions are unacceptable, since it is unrealistic to assume that compilers of a secret document would deliberately exaggerate its hidden treasure.⁵

1 Soon after the opening the smaller of the two copper rolls (Nov. 1956), Allegro suggested to Harding that ככ may be an abbreviation for ככרין (Allegro^{Pr}; Allegro, 37, etc., 135-36 n. 8; Allegro^{rev}, 14, 18, 20, etc.

2 Milik, 221 A-4, 287, etc.

3 Lurie, 76, etc.; Pixner, 346, etc.; Beyer, 224, etc.

4 The various weights are: (1) ככר, "talent" (sg.), once; ככרין, "talents" (pl.), 14 times. (2) ככ, 30 times, and כ once. (3) מִנִּין עֶשְׂרִין, "twenty minas" (12:9). מִנִּין, the plural of מִנָּה, "mina," is 1/60 of a ככר. It is known from Biblical, Mishnaic, and other sources. (4) עֲשָׂתוֹת זָהָב, "100 unused gold masses" (see #2, 1:5-6); עֲשָׂתוֹת זָהָב שֵׁשִׁין וְחֹמֶשׁ, "sixty-five unused gold masses" (2:4). The singular is עֶשֶׂת, its weight is unknown. (5) שֶׁבַע כֶּרִין אִיסְתֵּרִין (see #38, 9:2-3). אִיסְתֵּרִין is the plural of "stater," a Greek coin or weight about a quarter (or eighth) of a shekel.

5 Some accept the face value of the treasure listed in the Scroll, nevertheless, they consider it to be genuine; e.g., Pixner, 339-40. By comparing the treasures of the ancient world,

Those who consider the Scroll genuine document, have various opinions as to whom this massive wealth belonged. Among the suggestions offered are: the Qumran sectarians or Essenes, Zealots of Jerusalem, the assumed Temple of Bar Kokhba in Jerusalem, priestly shares collected after the destruction of the Temple, and the Temple itself.

The approach used in this study to resolve the critical issue of the veracity of the Scroll, is through comparative analysis of the three published drawings, the published photographs, some of the photographs of the Zuckermans, and the photographs obtained this author (see "Approach Used in this Study" in the Introduction), as well as examination of the units of weight in light of Biblical, Talmudic, Midrashic literature, and other sources.

It is accepted that ככ is an abbreviation for כֶּכָּרִין, "talents." This logical suggestion automatically drastically increases the wealth mentioned in the Scroll. For those who consider the document as fiction, this is helpful; larger wealth would indicate less trustworthiness. However, for those who accept the Scroll as genuine list, this may create a major problem.

It is difficult to calculate the actual weight of the כֶּכָּר (masc., pl. כֶּכָּרִין), "talent." In biblical times there were several types of measurements, regular, sacred or sanctuary (e.g., Exod. 38:24-26), royal (2 Sam. 14:26), and merchant (Gen. 23:16).⁶ While כֶּכָּר, "talent," occurs in the Masoretic Text and in the Targumim, it is rarely used in the Talmudic and Midrashic literatures.⁷

Both Talmudim relate a relevant incident. In the response to mockery by a Roman notable regarding the biblical calculation of מֵאָת כֶּכָּר, "one hundred talents" (Exod. 38:24; see Rashi, Rashbam, Ibn-Ezra, *Torah Temimah*), rather than the expected two hundred, Rabban Johanan ben Zaqnai replied that the sanctuary weights are twice the regular ones (y. San. 1:4 [7a], see *Penei Mošeh*, *Mar'eh haPanim*; b. Bekh. 5a, see Rabbeinu Gershom, Rashi, Tosaphot). This provides critical information. It shows that during the life of this Tanna, who witnessed the destruction of the Temple, sanctuary weights,

Harper concludes that 4630 talents of precious metal could have been part of the Jerusalem Temple treasury ("26 Tons of Gold and 65 Tons of Silver," *BAR* 19 [1993] 44-45, 70).

6 Cf. "Rabbi Haninah said, 'Whenever silver (shekels) are mentioned, in the Torah they are *sela's*, in the Prophets *litrae*, and in the Hagiographa *centenaria*, except the silver of Ephron (Gen. 23:16) ... which are *centenaria*'" (b. Bekh. 50a; cf. y. Qid. 1:3 [9b]).

7 כֶּכָּר, "talent," appears once in the Mishnah (m. Ter. 8:6), and four times in the Palestinian Talmud (M. Kosovsky, *Concordance to the Talmud Yerushalmi* 4, 599a). כֶּכָּר is neither mentioned in the Sifra, nor in the Mekhilta. כֶּכָּר in the Babylonian Talmud refers to: (1) in certain calculations connected with commandments (e.g., b. Bekh. 5a); (2) in connection with the Menorah (e.g., b. Men. 88b); and (3) as a hyperbolic expression (e.g., b. Suk. 51b; H.J. Kasowski and B. Kasowski, *Ozar Leshon haTalmud* 18, 223b-224a). כֶּכָּר, "talent," is also known from Ugaritic, Phoenician, Akkadian, etc. כֶּכָּר has additional meanings, such as כֶּכָּר לֶחֶם, "loaf of bread" (Jer. 37:21), כֶּכָּר הַיַּרְדֵּן, "Plain of the Jordan" (Gen. 13:10). It should be added that sanctuary talent occurs in the Dead Sea Scrolls (e.g., 4Q159).

especially the כֶּכָר, were no longer known. This implies that sanctuary weights ceased to exist by the end of the Temple era. Thus, one can assume that the Scroll refers to regular rather than sanctuary כֶּכָר.

The basic relevant unit is the שֶׁקֶל, "shekel" (from the root שָׁקַל, "to weigh"). There were שֶׁקֶל הָדוֹט, "common or regular shekel," and שֶׁקֶל קֹדֶשׁ, "sanctuary or sacred shekel" (Num. Rabbah 4:8). The Targumim and Talmudim equate the sanctuary shekel with סֵלָע, *sela*, as well as the Tyrian tetradrachm (Exod. 30:13; m. Bekh. 8:7). The regular כֶּכָר is equal to 3000 regular shekels, or 1500 *sela*'s, or 60 regular minas.⁸ Thus, when attempting to calculate the value of a כֶּכָר, one must know the weight of the shekel.

Weiss made detailed calculations of the ancient Jewish measurements and weights. His work is based on Biblical and Talmudical sources, on the opinions of the *ge'onim* and *poseqim*,⁹ as well as on the Hashmonean שֶׁקֶל יִשְׂרָאֵל, "Israel shekel." The latter, Weiss concludes, weighs 14.2 grams plus or minus one tenth of a gram.¹⁰ Other scholars arrived at somewhat different conclusions.¹¹

Based on the above, one can conclude that the Hashmonean shekel, the *sela*, the sanctuary shekel, the Tyrian tetradrachm, as well as two regular shekels, had essentially the same weight (c. 14.2 gram). Accordingly, a regular *kikkar* is about 21.3 kg; and a regular *mina* is about 3.55 kg.

כֶּכָרִין SCRIBAL CORRECTIONS REGARDING

There is agreement among the scholars that כֶּכָר is an abbreviation for כֶּכָרִין. Yet is this assumption valid? Can it be substantiated? The dilemma whether the Copper Scroll is a genuine document or not, essentially depends on the interpretation of the amount of wealth recorded in it. A careful examination of the three published drawings and the photographs point to a number of scribal corrections in the text. This can answer the question posed above. The term כֶּכָר, "talent(s)," appears fifteen times in the Scroll (see table below).

8 The difficult verse in Ezek. 45:12 indicates that the sacred mina (one sixtieth of a talent) which previously consisted of 50 shekels (cf. the Canaanite system) was increased to 60 shekels (cf. the Mesopotamian system), i.e., an increase by one sixth (see Targum, Rashi, Radak, *Mešudat David*, Malbim; b. Bekh. 50a, etc.). Accordingly, a sacred talent consisted of 3600 sacred shekels. See also, E. Stock, "Weights and Measures," *EJ* 16, 381-85.

9 Rabbis whose rulings were accepted as the definitive Halakhah.

10 Y. G. Weiss, *Midoth uMishkaloth shel Torah* (1984), 176.

11 E.g., A. Ben-David ("The Hebrew-Phoenician Cubit," *PEQ* 110 [1978], 27-28) weighted 940 Tyrian tetradrachmae and their average was 14.01 gr. According to H.P. Benish (*Middot weŠi'urei Torah* (1986/87), 423-24, 431, 433-34, 439-40, 444), the sacred weights were in biblical times: shekel 14.16 gram, mina 708 gr, talent 42.5 kg; and after the increase in the second Temple era: shekel 17 gram, mina 850 gr, talent 51 kg. The regular weights were half of these amounts.

THE TERM ככר IN THE SCROLL

1:15		[כ]כר	[כ]ספ ארבעין	# 5
1:4	KEN	שבעשרה	שדת כסף וכליה משקל	# 1
1:8		תשע מאת	נגד הפתח העליון	# 3
2:2	HN 42		בבור שתחת המעלות	# 6
2:6		שבעין	ב[ו] כלין וכסף	# 8
2:9	ΔI	עסר	בו כלין ובמוזקא שבו	# 9
2:15	22		חפור אמות בא ארבע	#11
7:7		ארבע מאות	אמות עסרין ו[אר]בע	#29
7:16		שתימ	עד הטור ככ 60 זהב	#32
10:7		ששין ושנין	לסמולו אמות עסר כסף	#46
10:10		שלש מאות זהב	אמות שתין הו הפתח	#47
12:1		ששין	תשע מאות זהב ככ 5	#56
12:3	42		כוזין תחת סף הכוך	#56
12:7		שש מאות	של דמע וכסף הכל	#58
12:9		71 מגין עסרין	בית הכוך הכל משקל	#59

For the proposed assumed original text before the amendment were made, see table below.

THE ASSUMED TEXT BEFORE SCRIBAL AMENDATION

1:15		[כ]כר	[כ]ספ ארבעין	# 5
1:4	KEN	שבעשרה	שדת כסף וכליה משקל	# 1
1:8		תשע מאת	נגד הפתח העליון	# 3
2:2	HN 42		בבור שתחת המעלות	# 6
2:6		שבעין	ב[ו] כלין וכסף	# 8
2:9	ΔI	עסר	בו כלין ובמוזקא שבו	# 9
2:15	22		חפור אמות בא ארבע	#11
7:7		ארבע מאות	אמות עסרין ו[אר]בע	#29
7:16		שתימ	עד הטור ככ 60 זהב	#32
10:7		כ	לסמולו אמות עסר כסף	#46
10:10		שלש מאות זהב	אמות שתין הו הפתח	#47
12:1		ככ	תשע מאות זהב ככ 5	#56
12:3	42		כוזין תחת סף הכוך	#56
12:7		ככ	של דמע וכסף הכל	#58
12:9	71	ככ	בית הכוך הכל משקל	#59

One can contrast the assumed pre-amended with the amended words (see table below).

PRE-AMENDED TEXT VS. AMENDED TEXT

	CORRECTED TEXT	ORIGINAL TEXT	B ^{dr} —M ^{dr}	A ^{dr}
1:4	ככרין	ככ	Y	Y
1:8		ככרין ?	N	?
2:2		ככרין	N	N
2:6	ככרין	ככ	Y	Y

2:9		ככרין	N	N
2:15		? ככרין	Y	N
7:7		ככרין	N	N
7:16		ככרין	N	N
10:7	כררין ששין ושנין	כ	Y	Y
10:10	ככרין	ככ	Y	Y
12:1	ככרין ששין	ככ	Y	Y
12:3	ככרין	ככ	Y	Y
12:7	ככרין שש מאות	ככ	Y	Y
12:9	ככרין מנין עשרין	ככ	Y	Y ¹²

The following is a detailed discussion of each occurrence of ככרין in the Copper Scroll.

- (1:4) Baker's drawing has only ככין, ככין, בכין, or בבין; Milik's has a small *reš-yod* ligature; while Allegro's a practically full sized *reš* in an angular position, plus a *yod* on the cut where the Scroll was sliced.¹³ The small or angular *reš* indicates an amendment by a second hand.
- (1:8) Although the *reš* is somewhat smaller than the preceding two relatively large sized *kafs*, apparently it is not an amendment.
- (2:2) The *reš* is smaller than the preceding two *kafs*. Allegro's drawing has a *yod-final nun* ligature. Probably it is not a scribal corrections.
- (2:6) Baker's drawing has only ככרין; Milik's has a *yod* of dotted line; while Allegro's has ככרין. The scribe wrote ככ and left out a full letter space, and later it was squeezed a small *reš*, plus a *yod* and a final *nun*. It is a an amendment.
- (2:9) Baker's drawing has only ככרין; Milik's a *yod* of dotted line; while Allegro's a clear *yod* running into a cut where the scroll was sliced. It is not a scribal correction. Allegro's drawing has more details than that of Baker and Milik. Since the *yod* is almost on the cut, it is missing from Baker's drawing, while Milik detected part of it.
- (2:15) In the drawings of Baker and Milik the letters רין is squeezed in between ככ and the numerals (the the head of the *reš* is narrow), but less so in Allegro's. According to drawings of Baker and Milik, it is an amendment, and perhaps so in Allegro's.
- (7:7) ככרין has a *reš-yod* ligature and a regular *nun* in final position. The ligature, as well as the *nun* in the drawings of Baker and Milik differ that of Allegro's. Probably it is not a scribal correction.
- (7:16) ככרין with a regular *nun* at final position. A crack can be seen in the drawing of Baker in the first *kaf*, but not in Milik's. On the other hand, a cut where the Scroll was sliced dissects between the two *kafs* in Allegro's drawing. It has no scribal correction.

12 The right two columns indicate, according to the drawings, whether there was a scribal correction. The abbreviations are: Y = yes; N = no.

13 Regarding letters on the cut, see "The Drawings" in the Discussion.

- (10:7) The drawings read כררין. The scribe left out at the beginning of the line about four letter size spaces, then engraved a single *kaf*. Later a second scribe, which can be seen from the script, added ררין, as well as ששין ושנין, a total of 13 letters. The two *rešes* are much smaller than the average letter of this line. There is a *lamed* in the next line below the second *reš*. The upper vertical line of this *lamed* goes well into the line above. One can see that the scribe was careful that the second *reš* should not touch the upper vertical line of the *lamed*. This indicates that the text was corrected after the *lamed* of the next line was engraved. Likewise, in Milik's drawing the final *nun* of כררין descends into the next line, and it is curled away from the *gimmel* below it, so it should not touch it. The amendment itself contains an error, it reads כררין instead of ככרין.¹⁴
- (10:10) The scribe originally wrote ככ and later רין was added. The rounded smaller size *reš* indicates the work of a second hand. It is an amendment.
- (12:1) The scribe originally wrote ככ left out some space, then wrote ביאתו. Later seven letters רין ששין were squeezed in. It is an amendment.
- (12:3) The scribe engraved ככ and later רין was added. The *reš* is a smaller size letter, perhaps the *yod* too. It is an amendment.
- (12:7) The first *kaf* of ככרין in Allegro's drawing has an unusual shape but not so in Baker's and Milik's. The *reš* is in an angular position and it is smaller than the usual letters of the line. In Allegro's drawing this *reš* is similar to the *reš* in line 1:4, while in Milik's to the small *reš* in line 12:9. In addition, the next two words, שש מאות, are written with smaller size letters, and differ from the letters of this column. This is an amendment.
- (12:9) The scribe engraved ככ, left out a good letter size space, then added the numerals. Later the letters רין were squeezed in between ככ and the numerals, and after the numerals the two words מנין עשרין were added. The *reš* and the *yod* are small size letters. Allegro's drawing has כיכרין with an extra *yod*, i.e., *plene* spelling. Allegro and Lurie do not indicate this extra *yod*.

Based on the shape of the *rešes* and some of the other letters, one can conclude that the corrections were made by a different scribe. All corrections have final *nuns*, either because it occupies less space than a regular one, or because the corrector was more careful.

These corrections may suggest: (1) ככ, and at least once כ, were intentionally written instead of ככרין so that if the original scribes understood the text, he should not be aware of the exact weight of material listed. (2) Later a more trusted second person added the additional information.

One can claim that ככ is the same as ככרין, and that the text had been corrected only at those places where the corrector found room for the corrections. However, an examination of the occurrences of ככ confirms the first observation. The abbreviation ככ occurs in the text 28 times. In about 14

14 For the possible readings of בררין for ברין, or כררין for כרין, see #46, 10:7.

cases there is more or less enough room to add the three letters ככרין. Thus, the scribe corrected only those cases which needed to be changed into ככרין.

The following table shows the abbreviations of ככ in the Scroll. It is organized according to unspecified ככ (1-16), silver ככ (17-25), silver and gold ככ (26), ככ and ככרין in the same line (27-28), and ככ followed by a lacuna (29-30) which can be restored as ככרין.

THE ABBREVIATION ככ IN THE COPPER SCROLL¹⁵

					B ^{dr} —M ^{dr}	A ^{dr}			
(1)	3:13	13	ככ		Y	Y			
(2)	5:14	32	ככ		N	N			
(3)	6:6	42	ככ		?	?			
(4)	6:10	21	ככ		Y	?			
(5)	6:13	27	ככ		N	N			
(6)	7:13	22	ככ		N	N			
(7)	8:9	4	ככ		Y	N			
(8)	8:13	66	ככ		N	N			
(9)	9:6	23.5	ככ		Y	Y			
(10)	9:9	22	ככ		Y	Y			
(11)	9:13	9	ככ		?	Y			
(12)	10:2	9	ככ		N	?			
(13)	10:4	12	ככ		N	N			
(14)	10:14	80	ככ		?	Y			
(15)	10:16	17	ככ		?	Y			
(16)	11:8	40	ככ		?	?			
(17)	4:10	מאתים	ככ	כסף	Y	Y			
(18)	4:12	שבעין	ככ	כסף	Y	Y			
(19)	4:4-5	55	ככ	כסף	?	?			
(20)	4:14	12	ככ	כסף	Y	Y			
(21)	5:4	7	ככ	כסף	N	N			
(22)	5:11	23	ככ	כסף	N	N			
(23)	8:16	70	ככ	כסף	N	?			
(24)	3:7	TP 40	ככ	כסף	Y	Y			
(25)	12:5	60	ככ	וכסף	N	?			
(26)	8:6-7	17	ככ	וזהב	כסף	N	N		
(27)	7:16	זהב	ככרין	שתיים	60	ככ	N	N	
(28)	12:1	ככרין	ששין	5	ככ	תשע מאות	זהב	N	N
(29)	7:2		[כ...]						
(30)	4:2	ΣK	[ככ...]						

One may ask: If ככ is an abbreviation for ככרין, why did the scribe engrave ככרין at all? Why did he need the extra three letters? Why did the scribe cor-

15 On the right side it is indicated whether there was space for amendment according to the drawings. The abbreviations are: Y = yes; N = no; ? = questionable.

rect some of the ככ's into ככרין and others not? Each character had to be hammered into the copper sheets with several hammer blows. Unnecessary characters meant extra work, time and expense. Moreover, since the Copper Scroll is a cryptic document, presumably the intention of the the compiler(s) was to create a text that would be difficult to comprehend, particularly concerning the wealth described in it. Additional abbreviations would make the text much more complex for those who should not read it.

The above questions lead to the next problem. If the correct reading at the bottom of the first column is [כ]ספ[א]רבעין [כ]כר, "(there are) forty [tal]lents of [sil]lver" (see #5, 1:14-15), then it raises several issues. If the singular collective ככר with plural numbers were sufficient as in Biblical Hebrew, why did the scribe use the plural ככרין at all? Why not only ככר? Why did the scribe correct ככ into ככרין by adding three letters when he could do it with only one, especially in those places where the space was limited. Moreover, if ככ is the same as ככר, with each abbreviation the scribe only gained one letter, namely a *reš*, not the three letters of רין.

These observations may also question the validity of reading ככר in the Scroll as mentioned earlier. Apparently, as in the Targumim, the singular ככר in the dialect of the compiler(s) was not used with numbers except of "one" (i.e., one talent). ככר could be an error for ככרין, or may be read as בכד, "in a pitcher," rather than ככר (see #5, 1:15). This also supports the suggestion that ככ is not an abbreviation for ככרין.

Item 56 is especially relevant to this discussion. It has both a small and large lacunae, scribal corrections, problems with words, phrase separations, subdivisions, and it is separated between columns. Lines 11:17 reads טיף על זהב ככ 5 ככרין ששין ביאתו מן הכרכ- [מ, ...] תשע מאות [המ]ע-רב. If the entire lacuna belongs to the first part of line 11:17, then תשע מאות refers to זהב, "nine hundred gold (pieces/shekels)." If a part of the lacuna belongs to the second part of the line, the last missing word(s) should define תשע מאות (such as כסף, "silver," כלין, "vessels," כדין, "pitchers," אמות, "cubits"), while ככרין ששין ביאתו is a separate phrase.

Line 12:1 contains seven letters squeezed in between ככ and ביאתו. The letters רי and שש are circa half sized; the two final *nuns* have nearly a regular shape, while the second *yod* is of normal size.¹⁶ Apparently, the scribe originally wrote ככ 5 זהב ככ, left a five square letters size space, then continued with ביאתו. This is not surprising, since elsewhere there are wide spaces between words (cf. 12:9). Later, judging from the shape of the letters, a different scribe corrected the text by squeezing in seven letters plus leaving two spaces for word separation. Accordingly, the text before correction had ככ 5 זהב ככ. תשע מאות זהב ככ. If ככ is an abbreviation for ככרין, then the scribe

¹⁶ This *yod* has an unusual shape in Milik's drawing.

could have simply added only ששין in full sized letters, and the text would read without any noticeable correction as תשע מאות זהב ככ 5 ככ ששין. Thus, there must have been a significant reason why the scribe felt compelled to squeeze in so many letters into a narrow space (see #56, 11:16-12:1).

ככרין occurs fourteen times in the Scroll. An intensive examination of the drawings reveals that the scribes wrote at least seven times ככ, that were later corrected into ככרין, and once כ, which was corrected erroneously into כרין instead of ככרין. In several cases additional information was added. At two locations one cannot conclude if there is a correction. Only four cases show no evidence of corrections being made in any of the three drawings.

Close study of the script indicates that several scribes engraved the original text.¹⁷ To maintain secrecy, probably scribes with limited Hebrew knowledge were employed. The corrections suggest that כ or ככ were intentionally written instead of ככרין, so that the scribes should remain unaware of the exact amount. Later, presumably a more trustworthy scribe added essential information. All corrections have final *nuns*, either because they need less space than regular ones, or it was revised by a more knowledgeable *editor*.

It can still be claimed that ככ is an abbreviation for ככרין, and that the text was corrected only at those places where the *editor* found room for amendments. However, examination of the occurrences of ככ indicates that at least in seven cases there is enough room to add or squeeze in the letters רין. Therefore, the text was corrected *only* where ככ had to be altered into ככרין.

Those who disagree with this view may argue that it is easier to use numerical symbols than to write out numbers, yet the scribes used both indiscriminately. Likewise, the scribes used ככ and ככרין. However, the fact that some ככs were corrected into ככרין and others not, shows that they are not interchangeable. Therefore, ככ is not an abbreviation for ככרין.

כֶּכֶר is the largest weight mentioned in the Masoretic Text and in the Rabbinic literature. In addition, ככרין and ככ are used in the Scroll in parallel fashion. These two facts demonstrate that ככ is a weight but less than a ככר.

The Elephantine Aramaic documents mention various weights, such as שקל, "shekel," the Persian כרש, "karsh" (equal to ten shekels), רבע, *reva*^c (quarter of a shekel), חלר, *hallur* (one tenth of a רבע). Often these weights are represented by their initials, ש, כ, ר, ח, respectively. Likewise, כסף, "silver shekel," is abbreviated as כש. In addition, the numerical symbols in the documents are practically identical with those used in the Scroll. E.g., 12 כרשן ... כסף 7 כרשן ... 6 ש 12 כ ... 8 שקלן 31 כרשן, "silver 31 karsh 8 shekel ... 12 k(arsh) 6 sh(ekels) ... 7 karsh silver ... 12 karsh" (Cowley, 22, 7:120-123); 2 כסף שקל, "2 silver shekels" (Cowley, 22, 1:1-3), versus 2 כש, "2 s(ilver) sh(ekels)" (Cowley, 22, 4:59-80); 10 כסף כרשן

17 This was already realized by Allegro^{Pr}.

לכרש 2 באבני מלכא כסף ר 2, “silver karsh 10 in royal weight, silver 2 q(arters) to one karsh” (Kraeling 4:20-21); ח 1 נחש דמי כסף שקל 1 ח 10, “1 bronze bowl which valued (in) silver at 1 shekel (and) 10 h(allurs) ... [1] cup [valued] (in) silver at 20 hallurs” (Kraeling 7:13-14).¹⁸

Some of these abbreviations and numerical symbols also occur among the Murabbaʿat texts. One of them (9:3) is a list of six lines. Each line starts with a כ followed by a numeral, then a ר, followed by a numeral, and once a מ, an abbreviation for מעאה, “obol,” a small coin. כ may be an abbreviation for כסף, “silver,” which often in the Masoretic Text means “silver shekel/piece” (see #1, 1:4; #59, 12:9). However, כ may be an abbreviation for כרש, as often in the Elephantine documents.

MURABBAʿAT TEXT 9:3

2	מ	8	ר	ז	כ	(1)
		1	ר	4	כ	(2)
		1	ר	1	כ	(3)
		4	ר	4	כ	(4)
2 ...	1	ר	5	כ	כ	(5)
		5	[ר]	5	כ	(6) ¹⁹

Similarly, a Phoenician inscription from Lapethos, Cyprus (c. 275 BCE) contains the phrase, 2 100 כר משקל [ב[ס]]²⁰. Donner and Röllig suggest that כר may be an abbreviation for the Persian weight כרש. Accordingly, it means, “in si[lv]er a weight of 100 and 2 kar(sh).”²⁰

Furthermore, a Hebrew inscription (c. seventh-sixth cent. BCE), in which every undamaged line except one out of 18, carries a personal name followed by a כ then by a numeral of 3, 4, 5, 7, or 8. E.g., 4 בן שחר לאחקם כ, “son

18 An Aramaic customs account from Egypt frequently uses the term כרש(ן). In addition, the genre of this account is very similar to that of the Copper Scroll. See A. Yardeni, “Maritime Trade and Royal Accountancy in an Erased Customs Account from 475 B.C.E. on the Ahiqar Scroll from Elephantine” in *BASOR* 293 (1994), 67-78; B. Porten and A. Yardeni, *Textbook of Aramaic Documents from Ancient Egypt* (1993), 3, xx-xxi, etc.

19 For clarification the lines are represented in columnar fashion. In the first line the waw after the kaf may indicate its numerical value “six”; in the second line one of the single numeral has a small circle above it. רבע שקל כסף is a “quarter” of a shekel, cf. רבע שקל כסף, “one quarter silver shekel” (1 Sam. 9:8); מעה (pl. מעות), in Aramaic מעין (pl. מעין), usually translated as “obol,” is a small coin, 1/24 of a סלע, sela^c (e.g., m. ‘Eruv. 7:11; Trg. 1 Sam. 2:36). In biblical times the ratio was one to twenty; e.g., עשרים גרה השקל, “the shekel is twenty gerahs,” is rendered by the Targumim as עשרין מעין סלעא, “the sela^c is twenty ma’a’s” (Num. 3:47). The six lines of the text can be translated as: (1) 6 k(arsh)/s(ilver shekel), 8 q(arters), 2 o(bols); (2) 4+ k(arsh)/s(ilver shekel), 1 q(arter); (3) 1 k(arsh)/s(ilver shekel), 1 q(arters); (4) 4 k(arsh)/s(ilver shekel), 4+ q(arters); (5) 5 k(arsh)/s(ilver shekel), 3+ q(arters); (6) 5 k(arsh)/s(ilver shekel), 5 [q(arters)].

20 *KAI* 1, 10; 2, 62 (#43:14).

of Shahaṛ for Ahiqam *k* 4";³ יועליהו בן עבדיהו כ, "Yo'aliyahu the son of 'Obadyahu *k* 3" (#[79]4, recto lines 8-9). It has been suggested that כ is an abbreviation for כסף, i.e., "silver shekel."²¹ Although it is unlikely, the כ may be an abbreviation for כרש.

Finally, the suggested vocalization of the various forms for כרש (prob. masc.), based on similar sounded masculine segolite nouns, are as follows.

In Hebrew: כָּרֶשׁ (sg.), כָּרָשִׁים (pl.), and כָּרְשֵׁי (pl. const.); cf. קָרֶשׁ, "board" (e.g., Exod. 6:18), קָרָשִׁים, "boards" (e.g., Exod. 26:22), קָרְשֵׁי, "boards of" (e.g., Exod. 26:17); כָּרֶם, "vineyard" (e.g., Exod. 22:4), כָּרְמֵי, "vineyard of" (e.g., Judg. 15:5), כָּרָמִים, "vineyards" (e.g., Deut. 6:11), כָּרְמֵי, "vineyards of" (e.g., Judg. 14:5).

In Aramaic: כָּרֶשׁ or כָּרֶשׁ (sg.), כָּרֶשָׁא (sg. emph.), כָּרְשֵׁין (pl.), כָּרְשֵׂא (pl. emph.), כָּרְשֵׁי (pl. const.); cf. כָּרֶם, "vineyard" (e.g., TO Exod. 22:4), כָּרְמָא, "the vineyard" (e.g., TO Gen. 9:20), כָּרְמִין, "vineyards" (e.g., TO Num. 16:14), כָּרְמֵא, "the vineyards" (e.g., TO Num. 22:24), כָּרְמֵי, "vine yards of" (e.g., Trg. Judg. 14:5). The plural in Egyptian Aramaic is כרשן (no *yod*), e.g., כסף כרשן, "silver karsh" (Cowley 5:7), which can be vocalized as כְּסֶפֶיכֶסֶף כְּרֶשֶׁן.

A SUGGESTED NEW MEANING FOR ככ

In light of the aforementioned discussion, it is very reasonable to propose that ככ is an abbreviation for כסף כרש(ין), "silver karsh," like כש for כסף שקל, "silver shekel," in the Elephantine documents, where כסף כרשן is commonly used (e.g., Cowley 5:7; Kraeling 9:20). One karsh is 10 shekels, about 71 gram. Since a talent consists of 3000 shekels, it is equivalent to 300 karsh. Based on the above, ככ represents a mere fraction of a talent.

The thirty cases of ככ can be classified into four groups: sixteen cases of unspecified ככ; nine cases of כסף ככ; one case of each of three phrases that differ; and two cases that are followed by a lacuna.

One may assume that ככ refers to כסף כרש(ין), "silver karsh." In those cases where כסף ככ is used, although one of the *kafs* seems to be redundant, i.e., כסף כסף כרש(ין), "silver silver karsh," it means the same. Similarly, in the Elephantine documents occasionally כסף is repeated. E.g., כל כספא ודמי, נכסיא כסף שקלן 7 חלרן 7 פלג, "all the silver and the value of the goods are (in) silver: silver 7 shekels, 7 (and) half hallurs" (Kraeling 2 verso 6-7).

Based on the above, three difficult phrases in the Scroll can be explained as follows.

כסף כרשין 60 זהב ככרין שתיים ככ 60 זהב ככרין שתיים, "60 silver karsh, (and) 2 talents of gold" (see #32, 7:16).

²¹ See R. Deutsch and M. Heltzer, *New Epigraphic Evidence from the Biblical Period* (1995), 92-102.

תשע מאות זהב כסף כרשין 5 ככרין means תשע מאות זהב ככ 5 ככרין ששין ששין, “nine hundred gold (pieces/shekels), 5 s(ilver) k(arsh), sixty (unspecified) talents” (see #56, 11:16-12:1).

17 ככרין כסף וזהב כסף means 17 ככרין כסף כרשין, “(an unspecified amount of) silver and gold, (plus) 17 s(ilver) k(arsh)”; or “one silver (piece/shekel) and one gold (piece/shekel), (plus) 17 s(ilver) k(arsh)” (see #34, 8:6-7).

Finally, it is possible that originally ככ was meant to be used for כסף כרשין, while later it was also used instead of כרשין alone. This may be the reason why nine times כסף, and possible twice זהב, precede ככ.

THE ABBREVIATION כ

The second half of line 11:7 reads, הו חרם כ. After this the three drawings differ. Baker's can be read as, 1 הו חרם כרש (= 1 karsh of consecrated silver,) or 11 הו חרם כ (= 11 karsh of consecrated matter.” Milik's and Allegro's can be read as, הו חרם ככ (= הוא חרם כסף כרש), “one karsh of consecrated silver,” or 10 הו חרם כ (= הוא חרם כרש), “10 karsh of consecrated matter” (see #52, 11:7).

If this case is read as ככ, it is an additional ככ situation in the Scroll. If it is read as a single כ, it could be an abbreviation for כרש, “k(arsh).” The latter is similar to the original reading of a single כ which was later corrected erroneously into ככרין ששין ושנין (instead of ככרין, “sixty-two talents,” (10:7). This further supports the concept of scribal corrections. Apparently the *editor* overlooked amending this site. Thus, there is a likelihood that the Scroll uses single *kaf* as an abbreviation for karsh.

One can ask: How come that an abbreviation for כרש, which is unknown in the entire Hebrew literature, would appear in the Scroll?²² In response, it should be noted that the Scroll does contain sets of Greek letters, and unusual words.²³ ככ may be another example of a unique element of this document.

The above discussion establishes that the scribes of the Elephantine documents, the Phoenician inscription, the Copper Scroll, the Murabba'at texts, etc., which extend across a time span of five hundred years, all used abbreviations and numerical symbols in a comparable fashion.

THE ESTIMATED WEIGHT OF THE LISTED SUBSTANCES

The following shows the estimated weight in talents listed in the Scroll according to Milik, Allegro, Lurie, and Pixner. The percentage depicts the portion of the total weight.

22 Essentially this question was raised at the *International Symposium on the Copper Scroll* (1996) at Manchester, England, by Dr. H. Eshel (Bar-Ilan University).

23 E.g., האסטנ (11:02), ואפורת, or ואפורת (1:9), טיף (11:07), הפרסטלון, or הפרסטלון (1:7), בשלף (8:10).

	MILIK ²⁴	ALLEGRO/LURIE ²⁵	ALLEGRO ^{rev}	PIXNER ²⁶
SILVER	2286 = 49.3%	3282 = 71.9%	3179 = 89.2%	---
SILVER/GOLD	2040 = 44.0%	---	---	4500
GOLD	307 = 6.7%	1280 = 28.1%	385 = 10.8%	---
TOTALS	4633 = 100.0%	4562 = 100.0%	3564 = 100.0%	4500

The next table is based on the author's suggested reading. The calculation assumes that: (a) the weights are regular, not sanctuary; (b) the weight of *ʿešet* is equal to a talent, although its weight is unknown; (c) ככ and כסף ככ refers to silver karsh; (d) where a specified amount of silver is mentioned without a given weight, it is assumed to be regular shekel; (e) the minimal weight of 4 unspecified *staters* is disregarded.

	TALENT	ʿESHET	MINA	KARSH	SHEKEL	KG	%
GOLD	302	165	---	---	900	9954	16.7%
SILVER	789	---	---	1059	---	16881	28.2%
UNSPEC.	1547	---	20	10	---	32959	55.1%
TOTALS	2638	165	20	1069	900	59794	100.0%

It is reasonable to assume that unspecified matter is silver, yet it may be copper. Given this possibility, the total wealth mentioned in the Scroll would be reduced even further. Although the total weight includes 165 *ʿešet*s as 165 talents of gold, it is still much less than those calculated by the scholars.²⁷

THE POSSIBLE READING OF בכדין

Since the letters *bet* and *kaf*, as well as *dalet* and *reš* are often indistinguishable in the Scroll, in seven to ten cases בכדין, "talents," can be read as בכדין = בכדין, "in pitchers." Likewise, כקר, "talent(s)," can be read as כקר, "in a pitcher" (1:15). Indeed, Allegro in line 7:16, and Beyer in line 10:7, read בכדין, "in pitchers." Interestingly, in only one or two cases, where the text was corrected, can it be read as בכדין.

If some of these are read as כקר or בכדין, it would drastically reduce the wealth listed in the Scroll. E.g., "forty talents of silver" (= c. 852 kg), vs. "forty silver (pieces/shekels) in a pitcher" (= c. 0.57 kg), a reducing ratio of 1:1500 (see #5, 1:15).

24 Milik (282) considers unspecified material as a mixture of silver and gold (see his table).

25 Allegro (58-59, 135-36 n. 8; Allegro^{rev}, 42) includes in "silver" mixed gold and silver plus unspecified material. Lurie accepts Allegro's calculations.

26 Pixner (339-40) accepts the total sum of c. 4500 talents of silver and gold.

27 Although the actual weight of *ʿešet* is unknown, one may assume that it might have been a *kikkar*. Cf. Rashi on Num. 8:4 explains that the Menorah was beaten out from עשת של ככר, "a gold mass of a talent."

IS THERE A SCRIBAL AMENDATION REGARDING ככרין ?
CAN בכרין BE READ AS ככרין ?²⁸

	WAS THE TEXT CORRECTED ?		ORIGINAL READING		CAN BE READ AS בכרין ?	
	B ^{dr} —M ^{dr}	A ^{dr}			B ^{dr} —M ^{dr}	A ^{dr}
1:4	Y	Y	ככרין	ככ	N	?
1:8	N	?		ככרין ?	Y	Y
2:2	N	N		ככרין	Y	Y
2:6	Y	Y	ככרין	ככ	Y	Y
2:9	N	N		ככרין	Y	Y
2:15	Y	N		ככרין ?	Y	Y
7:7	N	N		ככרינ	Y	Y
7:16	N	N		ככרינ	Y	Y
10:7	Y	Y	כרין	כ	N	N
10:10	Y	Y	ככרין	ככ	N	N
12:1	Y	Y	ככרין	ככ	?	Y
12:3	Y	Y	ככרין	ככ	?	?
12:7	Y	Y	ככרין	ככ	N	N
12:9	Y	Y	ככרין	ככ	N	N

The following is a list of cases where at least in one of the three drawings it is possible to read בכרין. They are grouped in five groups. Although these readings drastically reduce the wealth listed in the Scroll, they may make the text more secretive.

(1:14-15) ב[כ]סף ארבעין [ב]כר, "forty [si]lver (pieces/shekels) [in] a pitcher."

(2:6) וּכְסָף בְּכַדִּין שְׁבַעִין, "and (an unspecified amount of) silver (are) in seventy pitchers," or "seventy silver (pieces/shekels are) in (an unspecified number of) pitchers."

(12:1) וְזָהָב בְּכַדִּין שְׁתֵּימָה, "(an unspecified amount of) gold (are) in two pitchers," or "two gold (pieces/shekels are) in (an unspecified number of) pitchers." These three readings resemble each other.

(7:7) 22 בְּכַדִּין, "(an undefined number of unspecified objects are) in 22 pitchers," or "in (an undefined number of) pitchers (are) 22 (unspecified objects)."

(12:3) 42 בְּכַדִּין, "(an undefined number of unspecified objects are) in 42 pitchers," or "in (an undefined number of) pitchers (are) 42 (unspecified objects)."

(12:1) 60 בְּכַדִּין שְׁשִׁין, "(an undefined number of unspecified objects are) in sixty pitchers," or "in (an undefined number of) pitchers (are) sixty (unspecified objects)."

28 In line 10:7 the corrected text reads כרין instead of ככרין. In Allegro's drawing line 12:9 has כיכרין. The phrase, ב[כ]סף ארבעין [ב]כר, "forty silver (pieces/shekels) in a pitcher," can be read as, ב[כ]סף ארבעין [כ]כר. Abbreviations: Y = yes; N = no; ? = questionable.

- (7:16) בְּכַדִּין אַרְבַּע מֵאוֹת, “(an undefined number of unspecified objects are) in four hundred pitchers,” or “in (an undefined number of) pitchers (are) four hundred (unspecified objects).”
- (1:8) בְּכַדִּין תִּשְׁעַ מֵאוֹת, “(an undefined number of unspecified objects) in nine hundred pitchers,” or “in (an undefined number of) pitchers (are) nine hundred (unspecified objects).”
- (2:2) HN 42 בְּכַדִּין, “(an undefined number of unspecified objects are) in 42 pitchers. (Code: HN),” or “in (undefined number of) pitchers (are) 42 (unspecified objects). (Code: HN).” Perhaps HN defines the content.
- (2:9) ΔΙ 10 בְּכַדִּין עֶסֶר, “(an undefined number of unspecified objects are) in ten pitchers. (Code: ΔI),” or “in (undefined number of) pitchers (are) 10 (unspecified objects). (Code: ΔI).” Perhaps ΔI defines the content.
- (1:3-4) KEN כֶּסֶף וְכִלְיָהּ מְשָׁקֶל בְּכַדִּין שִׁבְעָשָׂר, “one silver carrying chair and its implements, (plus) seventeen weight(s) in (an unspecified number of) pitchers. (Code:) KEN.”

OTHER POSSIBLE INTERPRETATIONS FOR ככ

Since often in the Scroll *kaf* and *bet* are indistinguishable, ככ can also be read as בכ, כב, and בב. Although these seem to be meaningless, some Aramaic words are close to these possibilities.

כב = כוב or כב = כוב — The Hebrew כוּבוֹב (pl. כוּבוֹיִם; e.g., Gen. Rabbah 49), “thorn,” seems to be an Aramaic loanword. In Aramaic כּוּבָא (pl. כּוּבָיִן, emph. כּוּבָיָא, const. כּוּבִי) has several additional meanings, such as “pitcher,” e.g., כּוּבָא מְלֵיא, “a full pitcher” (b. ‘A.Z. 60a), cf. שְׁנֵי דוּדֵי-כֶסֶף, “two pots full of silver,” in the Scroll (4:8), and a sort of cake, e.g., כּוּבָא דְאַרְעָא, “cake baked on the ground” (b. Ber. 37b). One can assume that in Hebrew כּוּבוֹב also means “pitcher” and “cake.”

ככ = כוף or ככ = כוף, “cake” — The above phrase also occurs as כּוּבָא דְאַרְעָא, “cake baked on the ground” (e.g., b. Git. 68a), which may be only a graphical letter interchange of *bet-kaf*. This possibility should not be confused with כּוּף־כּוּף, “burial niche,” which occurs in the Scroll (12:3, 12:8).

ככ = כף may be a Hebrew form of the Aramaic כְּכָא, “tooth, molar,” a synonym of שִׁנָּא (= הֶשֶׁן in Hebrew), “(the) tooth.” E.g., כְּכִיָּה וְשִׁנֵּיהָ, “his molar(s) and his tooth/teeth (of) his mouth” (PsJ Num. 21:35).²⁹ כְּכָא, as an alternate reading occurs in the Scroll, שֶׁל הַכְּכָא שֶׁל הָרֶגֶב, “of the molar of the clod,” i.e., “of the peak of the clod” (see #26, 6:7-8).

ככ may be a shortened form of כּוּכָב (masc., pl. כּוּכָבִים), “star,” where originally the *waw* was consonantal, something like כּוּכָב or even כּכְכָב, a reduplicated form of ככ; cf. the Arabic *kawkab* and Ugaritic *kbkb*.³⁰

²⁹ Jastrow (638a) suggests that ככא is a collective noun, i.e., “molars.”

³⁰ Three plural forms are attested in Ugaritic: *kbbm*, *kbkbm*, and *kbkbt* (UT, 417b [#1189]; WUS, 145 [#1277]).

The possible reading, כב = כוב or כב = כוב, "cake," ככ = כוך or ככ = כוך, "cake," כך = כך, "molar," and כב = כוכב, "star," can be explained as certain ornaments; cf. the biblical כפתור ופרח, "a knob and a flower" of the Menorah (Exod. 25:33). Also, they could be nicknames for weights.

Moreover, one of the meanings of כָּכָר is "loaf"; e.g., וְכָכָר לֶחֶם אֶחָת, "and one loaf of bread" (Exod. 29:23). כוֹף, כוב, כוב, or כוֹף, "cake," are synonyms of כָּכָר, "loaf," and they may be semantically similar. Since כָּכָר means bot "loaf" and "talent," so it is possible that כב, ככ, or ככ, also denotes a weight or coin of a smaller size. Thus ככ, ככ, or ככ may represent a small weight or coin.

However, כב, ככ, ככ, and ככ are singular, and one would expect plural forms, i.e., ככין = כוכין and ככין = כוכין; cf. ככרין שבעין (2:6). Yet in the phrase ככרין ארבעין [כ]כרין, "forty [tal]lents/talents of [si]lver," the singular ככר is used as a collective noun (see #5, 1:14-15).³¹

Furthermore, singular collective also occurs with numbers (often preceding it as in the Scroll). E.g., וּבְכָר שֶׁשָּׁה וּשְׁלִשִּׁים אֶלֶף, vs. וְתוֹרֵי תְלָתִין וְשָׁמָא אֶלֶפִין (TO), "and thirty-six thousand oxen" (Num. 31:44).

Based on the above discussion, in all 30 cases of ככ in the Scroll these solutions would be meaningful, as the following tables demonstrates.

IF THE READING IS כב (OR ככ) IN THE SCROLL

3:13	כב 13	13	pitchers
4:2	ככ [...]	[...]	pitchers ככ
5:14	ככ 32	32	pitchers
6:6	ככ 42	42	pitchers
6:10	ככ 21	41	pitchers
6:13	ככ 27	27	pitchers
7:2	ככ [...]	[...]	pit[chers]
7:13	ככ 22	22	pitchers
8:9	ככ 4	4	pitchers
8:13	ככ 66	66	pitchers

31 Many nouns occur in singular (as a collective), and in plural. E.g., ויהי לי שור וחמור, literally "And I (Jacob) have acquired ox, and donkey, flock, servant, and maid servant," but the proper meaning is as Onkelos renders, ויהו לי תורין וחמורין עאן עבדן, "And I have acquired oxen, and donkeys, flocks, servants, and maid servants" (Gen. 32:6). In a similar verse most of these nouns are in the plural forms, ויהי לו עאן ודקר וחמורים, and its Targum Onkelos, ויעבדים ושפחות ואתנת וגמלים ויהו ליה עאן ותורין וחמורים ועבדן ואמקן, "and he (Abraham) acquired flocks, herds (or oxen), donkeys, slaves, and maid servant, female donkeys, and camels" (Gen. 12:16). The rest of the collective singulars also occur in plural forms; e.g., שוררים (masc.), "oxen" (Hos. 12:12), בקרים (masc.), "herds, oxen" (e.g., Amos 6:12; 2 Chron. 4:3), and the plural of צאן (fem.), "flock," in the verses, ואת בכורי בקרינו וצאנינו, "and the firstlings of our herds and of our flocks" (Neh. 10:37), and probably ונלך בבנינו ובבנותנו בצאנינו ובבקרנו נלך, "We will go with our young people and with our old people, we will go with our sons and with our daughters, with our flocks and with our herds" (Exod. 10:9).

9:6	כב 23.5	23.5	pitchers
9:9	כב 22	22	pitchers
9:13	כב 9	9	pitchers
10:2	כב 9	9	pitchers
10:4	כב 12	12	pitchers
10:14	כב 80	80	pitchers
10:16	כב 17	17	pitchers
11:8	כב 40	40	pitchers
4:10	כספ כב מאתים	two hundred silver	pitchers
4:12	כספ כב שבעינ	seventy silver	pitchers
3:7	כספ כב 40 TP	49 silver	pitchers TP
4:4-5	כסף כב 55	55 silver	pitchers
4:14	כסף כב 12	12 silver	pitchers
5:4	כסף כב 7	7 silver	pitchers
5:11	כסף כב 23	23 silver	pitchers
8:16	כסף כב 70	70 silver	pitchers
12:5	וכסף כב 60	and 60 silver	pitchers
8:6-7	כסף וזהב כב 17	17 silver and gold	pitchers
7:16	כב 60 זהב ככרינ שתים	60 pitchers, two gold talents	
12:1	כב 5 ככרין ששין זהב	5 gold pitchers, sixty talents	

These are similar to the phrases כַּדִּין שֶׁל כֶּסֶף שֵׁשׁ, “six silver pitchers” (2:11, 7:10), and שִׁבְעָה כַּדִּין אִיסְתָּרִין אַרְבַּע, “seven pitchers (and) four *staters*” (9:3). If it is rendered as “cakes,” then “pitchers” must be substituted by “cakes.” Both vocalization כב and כָּב are appropriate.

IF THE READING IS כָּב IN THE SCROLL

3:13	כָּב 13	13	molars
4:2	כָּב [...] ΣK	[...]	molars ΣK
5:14	כָּב 32	32	molars
6:6	כָּב 42	42	molars
6:10	כָּב 21	41	molars
6:13	כָּב 27	27	molars
7:2	כָּב [...] pit[chers]	[...]	pit[chers]
7:13	כָּב 22	22	molars
8:9	כָּב 4	4	molars
8:13	כָּב 66	66	molars
9:6	כָּב 23.5	23.5	molars
9:9	כָּב 22	22	molars
9:13	כָּב 9	9	molars
10:2	כָּב 9	9	molars
10:4	כָּב 12	12	molars
10:14	כָּב 80	80	molars
10:16	כָּב 17	17	molars

11:8	כֶּכֶּ 40	40	molars
4:10	כֶּסֶּפְּ כֶּכֶּ מֵאָתַיִם	two hundred	silver molars
4:12	כֶּסֶּפְּ כֶּכֶּ שִׁבְעִינָן	seventy	silver molars
3:7	TP 40 כֶּכֶּ	49	silver molars TP
4:4-5	כֶּסֶּף כֶּכֶּ 55	55	silver molars
4:14	כֶּסֶּפְּ כֶּכֶּ 12	12	silver molars
5:4	כֶּסֶּפְּ כֶּכֶּ 7	7	silver molars
5:11	כֶּסֶּפְּ כֶּכֶּ 23	23	silver molars
8:16	כֶּסֶּף כֶּכֶּ 70	70	silver molars
12:5	כֶּסֶּף וְכֶכֶּ 60	and 60	silver molars
8:6-7	כֶּסֶּף וְזָהָב כֶּכֶּ 17	17	silver and gold molars
7:16	כֶּכֶּ 60 זָהָב כִּכְרִינִי שְׁתֵּימָה	60	molars, two gold talents
12:1	כֶּכֶּ 5 כִּכְרִינִי שְׁשִׁין זָהָב	5	gold molars, sixty talents

If the text ככ is read as כב, a shortened form of כוכב, then it must be substituted, and rendered as "stars."

CONCLUSION

The above analysis strongly challenges the accepted view that ככ is an abbreviation for ככר(ין). It strongly suggests that:

- (1) The original text was written by more than one scribe, and was corrected by other scribes, who often amended the original כ or ככ into ככרין.
- (2) ככ is clearly not an abbreviation for ככרין, "talents."
- (3) ככ most likely is an abbreviation for כסף כרש(ין), "silver karsh." Karsh, a Persian weight equal to 10 shekels, is a minute part of a talent of 3000 shekels.

The conclusion that ככ is only a mere fraction of a talent has significant ramifications. Many scholars cannot accept the Copper Scroll as a genuine document on account of the presumably massive wealth recorded in it. The 'official' announcement at the opening and preliminary reading of the Scroll refers to 200 tons of gold and silver. This study revises that grossly exaggerated amount to less than 60 tons of precious metal, of which less than 17% is gold, the rest being silver and unspecified metal that could be copper or other metals. The aforementioned new interpretation strongly supports the view of those who accept the Copper Scroll as a historically valid document. If this is the case, then the status of the Copper Scroll among the Dead Sea Scrolls is significantly enhanced.

APPENDIX B

NUMERALS, NUMBERS, AND CONTRACTED TEENS

INTRODUCTION

The Copper Scroll consists of circa 713 words (including those partially restored), 31 sets of numerical symbols (i.e., numerals), and seven sets of two-three Greek letters (see Appendix C). In addition, there are lacunae that cannot be reconstructed. About 16 percent of the combined word-numeral count of the Scroll are numerals and cardinal numbers.

NUMERALS AND NUMBERS

The numerals used in the Scroll are known from various sources, such as Dead Sea Scrolls (e.g., 4Q320; 4Q554; 4Q559), Murabba'at documents (e.g., 8; 9), Phoenician (e.g., *KAI* 1, 3,7 [#14:1; #33:1]), Ancient Aramaic (e.g., *KAI* 1, 45,49 [#228:1; #250:1-2]), Elephantine Aramaic documents (e.g., Cowley 5; 36), funeral inscriptions from the Second Temple period (e.g., Bethphage), etc.¹

The numerals in the Scroll range in dimension from 4 to 100. The symbols are: 0.5 (in 23.5), which looks like a final *pe*, ף (may be an abbreviation for the Aramaic פֿלֵג, “half”; see #39, 9:6); vertical lines for units (often connected at their bottoms); 10; 20 (created by two tens, one on top the other); and 100. Numerals in the Scroll occur only after the abbreviation ככ, כִּכְרִין, “talents,” and עֲשָׂתוֹת, “bars, masses” (see #2, 1:5; Appendix A). They are always at the end of an item, except once, where the text originally ended with numerals, and later two words were added (see Appendix A). Also, five times numbers or numerals are followed by Greek letters (see Appendix C).

There is a total of 83 cardinals in the Scroll. They can be subdivided into thirty-four “units,” nine “teens,” eight “tens,” six “compound tens,” six “hundreds,” and one “compound hundred.”

In Hebrew the cardinals are expressed in units, teens, hundreds, thousands, and ten thousands. The cardinals are substantives, except אֶחָד (masc.) and אֶחָת (fem.), “one,” which are adjectives.² The suffix *-ayim*, as in many

1 See G.B. Sarfati, “מִסְפָּר” in *EB* 5, 170-85. Sarfati lists numerals from Phoenician (173); Elephantine documents (174); Palestinian funeral inscriptions from the late Second Temple period (176); Copper Scroll (177); and a jar from Qumran (177). Concerning the Qumran material, see P. Benoit, J.T. Milik, and R. de Vaux, *DJD II*, 88-91.

2 Some scholars also consider “two” as an adjective; cf. M.Z. Segal, *Diqduq Lešon ham-*

nouns, indicates duality: שְׁנַיִם (masc.) and שְׁתַּיִם (fem.), “two,” מֵאָתַיִם, “two hundred” (i.e., twice מֵאָה, “hundred”), אֲלָפַיִם, “two thousand” (i.e., twice אֶלֶף, “thousand”), and רִבְבוֹתַיִם, “twenty thousand” (i.e., twice רִבְבָּה, “ten thousand”).³ Regarding שְׁנַיִם and שְׁתַּיִם, see #38, 9:2.

In the Scroll often the spelling of numbers deviate from the Hebrew *norm*, such as *samekh* for *šin*, *ʾalef* for *he*, and *nun* for *mem*. Such variations can be found in the Bar-Kokhba letters, and in better Mishnah manuscripts. These features are common in Aramaic.

In addition, there are a few ordinals: שְׁנִי, “second (masc.)” (1:11), שְׁנִיָּא, “second (fem.)” (9:4 and 10:1, respectively), שְׁלִישִׁי and שְׁלִישִׁית, “third (masc.)” (2:4 and 1:5, respectively), שִׁבְעִי וְשִׁבְעִי, “seventh (masc.)” (1:10, alt.); two fractions: מֶחֶצֶא, “half (fem.)” (9:6), מֶעֶשֶׂר, “tenth, tithe (masc.)” (1:12); plus terms like שָׁבַע וְשָׁבוּעַ, “week year, seventh year (masc.)” (1:10), מִשְׁנָה, “second, copy (masc.)” (12:11), אֲחֵרָה, “other, second (fem.)” (3:5), כָּל, “all, every” (3:4, 9:16, 12:5, 12:12), and הַכֹּל, “the total” (1:10, 12:7, 12:9).

COMPOUND NUMBERS

The compound numbers (except teens) have three orders in Biblical Hebrew: ascending, descending, and a combination of both. The Targumim and Mishnaic Hebrew use the descending order. To visualize that, the examples are translated literally. חֲמִשָּׁה וְשִׁשִּׁים וְשָׁלֹשׁ מֵאוֹת וְאֶלֶף, “five and sixty and three hundreds and one thousand,” vs. אֶלֶף וְתֵלַת מֵאָה וְשִׁתִּין וְחֲמִשָּׁה, “one thousand and six hundred and sixty and five” (Num. 3:50); שָׁנָה וְעֶשְׂרִים, “one hundred year and twenty year and seven years,” vs.

Mišnah (1936/37), 100; Moscati, 116.

עֶשְׂרִים, “twenty,” may belong to this group, perhaps originally it was עֶשְׂרִים or עֶשְׂרִים, i.e., twice עָשָׂר, “ten” (cf. Ibn-Ezra, *Sefer hamMispar*; *Sefer Sahut*; Moscati, 117). Others consider עֶשְׂרִים as the plural of עָשָׂר, like שְׁלִישִׁים, “thirty,” of שָׁלֹשׁ, “three,” etc. (cf. BDB, 797b; SB, 672a; ESh^{dic}, 2019b; H.L. Stark, *Hebräische Grammatik mit Übungsbuch* [1902], 50; M. Rath, *Sefat Amenu* [1920], 97; A. Kahana, *Diqduq hallašon haʿIvrit* [1931], 110; T.O. Lambdin, *Introduction to Biblical Hebrew* [1971], 219). Accordingly, עֶשְׂרִים would mean “one hundred,” while “twenty” would be formed as the plural of שְׁנַיִם, “two,” which itself is a dual formation. Moreover, אֶחָד, “one” (Gen. 11:1; Ezek. 37:17), “few, several” (Gen. 27:44, 29:20; Dan. 11:20), the plural of אֶחָד, should mean “ten.” Hetzron suggests that not only עֶשְׂרִים, but even שְׁלִישִׁים, אֶרְבָּעִים, etc., were originally dual formations (“Innovations in Semitic Numeral System,” *JSS* 12 [1977], 191). Accordingly, שְׁלִישִׁים should mean “six,” אֶרְבָּעִים, “eight,” etc., and אֶחָד, “two.” Therefore, except עֶשְׂרִים and אֶחָד, the suffix *-im/-in* from thirty to ninety denotes *tenfoldness*; e.g., שָׁלֹשִׁים means ten times שָׁלֹשׁ, i.e., “thirty.” It should be added that many abstract nouns are formed with the suffix *-im/-in*, e.g., נְעֻרִים, “youth” (Ps. 127:4); זְקֵנִים and its Targum סִיבְתִּין, “old age” (Gen. 44:20); צְלוּמִים, “youth” (Ps. 89:46). Harris says that in Phoenician and other Semitic languages, some nouns are plural in form but not in meaning, cf. חַיִּים = חַיִּים, “life” (*A Grammar of the Phoenician Language* [1936], 60).

In the Scroll, the compound numbers (except teens) are in descending order. E.g., כָּל שֵׁשׁ מֵאוֹת וְחָמֶשׁ (fem.), “a total of six hundred and nine” (3:4), אֲמוֹת שְׁמוֹנָה וּמֵחָצֵא (masc.), “sixty-two talents” (10:7); אֲמוֹת שְׁמוֹנָה וּמֵחָצֵא (fem.), “eight and half cubits” (9:5-6). There is one exception. The suggested reading of אֲמוֹת אַחַת אַרְבָּעִין, “forty-one (fem.) cubits,” is asyndetic, and the connective “and” is absent. This may be a dialectical peculiarity. This phenomenon may occur once in the Masoretic Text, but it is known from Phoenician, Punic, and Neo-Punic inscriptions (see #1, 1:3).

7 Allegro, 33, 35, 134 n. 7, 140 n. 48; Allegro^{rev}, 21-22.

עסרא, "fifteen," rather than תשעסרא, "nineteen" (with a *taw*).⁸ LURIE: שבועשרה (= שבע עשרה); and תשע סרא (= תשע עשרה).⁹ WOLTERS and BEYERS: שלוש <עש>רא; שבועשרה; and תשעסרא.¹⁰ PIXNER, GARCIA MARTINEZ, and VERMES agree with Milik, while WISE follows Lurie.¹¹

These three cases seem to be contracted teens. These are unknown from other Hebrew sources, but used in various Aramaic dialects (and Akkadian).

תשע עשרה = תשעסרא (fem.), "nineteen," where the two 'ayins merged, *samekh* instead of *sin*, and 'alef instead of *he* (2:8). In the drawings of Baker and Milik תשעסרא is a single word having some spaces before and after. In Allegro's drawing תשע is followed by a space, then a vertical dotted line passes through the column where the Scroll was cut open. This cut removed the left side of the *samekh* (see "The Drawings" in the Discussion). When the drawings of the two halves of the Column 2 were matched, the left half of lines 7-8 became lower than the right side. As a result, תשע is about 5 mm lower than סרא. Accordingly, the reading is תשע סרא, a non-contracted teen, as often in Aramaic. Yet this does not necessary mean that it is not Hebrew. Moreover, often in the Scroll there is a space between the letters within a word, thus it may be considered as one word (e.g., in line 3:13 between the two *kafs* of ככ there is a space for at for another *kaf*).

שבוע עשרה = שבועסרא (fem.), "seventeen," where the two 'ayins merged (1:4). Since the Scroll is often in continuous script, שבועסרא can be separated as שבע עשרה, where the 'ayin of עשרה disappeared, as in some Aramaic dialects. However, there is a three letter size space between שבועסרא and the following Greek letter, while the 'ayin and *sin* touching each other. Also, the last three characters of the preceding ככרין is a later scribal addition. Originally there was enough space between ככ and שבועסרא. This can be better recognized in Milik's drawing than that of Allegro.¹² Therefore, one can assume that this is a contracted teen.

שלוש עשרה or שלושרא for שלוש עשרה, where the 'ayin of עשרה is deleted, *sin* and *sin* merged, plus 'alef instead of a *he* (9:2). Since שלושרה has only one ש, it cannot be separated into two words, unless it is amended to שלוש עשרה. As the first part of שלושרה indicates, it is a Hebrew formation (שלוש in Aramaic is תלת).

Judeo-Aramaic contracted teens are known from Megillat Ta'anit,¹³ Pseudo-Jonathan,¹⁴ Targum Chronicles,¹⁵ and the Babylonian Talmud.¹⁶

8 Milik, 284, 286, 293, 229 B-10a.

9 Lurie, 53, 56 n. 7, 58, 70-71 and n. 2.

10 Wolters, 32-35, 48-49; Beyer, 225-26, 229-30.

11 Pixner, 342, 346, 352; Garcia Martinez, 261-62; Vermes, 374, 376; Wise, 191-92, 195.

12 Baker's drawing has ככיןשב שרה (no *reš*, nor 'ayin). Regarding scribal corrections of the term ככרין, see Appendix A.

13 Regarding Megillat Ta'anit, see "Hebrew of the Copper Scroll" in the Discussion.

	MEGILLAT TA'ANIT	PSEUDO- JONATHAN		TARGUM CHRONICLES		BABYLONIAN TALMUD	
#	masc.	masc.	fem.	masc.	fem.	masc.	fem.
11		חדסר	חדיסרי	חדסר	חד סרי	חדסר חדיסר	
12		תריסר (תרתיסר)	תרתיסרי	תריסר		תריסר (תליסר)	
13		תליסר	תיליסרי	תליסר תלתסר		תליסר	תליסרי (תליסר)
14	ארביסר	ארביסר	ארבסרי	ארבסר	ארביסרי	ארבסר ארביסר	ארבסרי (ארביסר) (ארבעיסר)
15	חמיסר	חמיסר	חמיסרי	חמיסר		חמיסר	חמסרי
16		שיתסר שיתיסר		שיתסר	שיתסרי	שיתסר	שיתסרי (שיתסר)
17	שיבסר		שבסרי שביסרי	שבסר שיבסר		שבסר שיבסר	שבסרי
18		תמניסר		תמניסר	תמניסרי	תמניסר	
19			תשסרי	תשסר (תשיסרי)		תשסר	תשסרי תשיסרי תישסרי

These four Judeo-Aramaic sources complement each other. Many of the contracted teens might originally have been non-contracted, or vis-à-vis. The feminine forms end with a *yod*. Probably, due to the weakening of the gutturals the 'ayins of עסר, עסרי, שבע, and תשע were lost. Some of these have variant spellings. An extra *yod* may facilitate the pronunciation.

Syriac (Christian Aramaic) is closely related to Judeo-Aramaic.¹⁷ In Syriac

14 תרתיסר (masc.) seems to be a scribal error for תריסר.

15 Often in the Targum of Chronicles ordinals (frequently prefixed with a *lamed*) are used instead of cardinals. In this list, חדסר, תליסר, ארבסר, חמיסר, שיתסר, שיבסר, תשיסר, and תשיסרי are from such cases (1 Chron. 25:20-26). תשיסרי is a feminine form used instead of the expected תשיסר, it may be a scribal error or a variant of the cardinal תשיסרא "nineteenth." Perhaps חר סרי (fem., 2 Chron. 36:5,11) would indicate a combined form of חרסר. Most of these are from A. Rahmer, *Targum šel Divrei hayYamim* (1866).

16 The Babylonian Talmudic Aramaic frequently uses masculine for feminine forms. These are indicated in parenthesis. Often תליסר means תריסר, "twelve," a *reš-lamed* interchange (J. Rosenberg, "Das aramäische Zahlwort im babylonischen Talmud," *MGWJ* NF 18 (1910), 400-04; J.N. Epstein, *A Grammar of Babylonian Aramaic* [Heb., 1960], 125-27). תליסר may be Syriac where the *Ayin* is retained. תריסר and תליסר also occurs in the Aramaic Zohar (M.Z. Kaddari, *The Grammar of the Aramaic of the "Zohar"* [1971], 62).

17 According to Kutscher, the language of the Palestinian Talmud and aggadic Midrashim is Galilean Aramaic, while the Palestinian Targum and Pseudo-Jonathan is practically Galilean Aramaic ("Aramaic," *EJ* 3, 270; "אַרְמִית" in *EH* 5, 965). Levitas says that in Galilean Aramaic "units" and "ten" are not connected (*A Grammar of the Galilean Aramaic* [1986], 219). Kutscher discusses the numbers in Exodus Rabbah, but he does not mention

the teens are contracted, yet the 'ayin of (א)עסר is retained. In addition, the feminine forms end with an 'alef, some have a *taw* at the end of the unit before the 'ayin of (א)עסר, and some retain the 'ayin of ארבע and שבע.¹⁸ Contracted teens are also used in Mandaic (an Aramaic language).¹⁹ In Akkadian only four teens are attested; they happened to be contracted.

#	SYRIAC				AKKADIAN	
	masc.	masc.	fem.	fem.	masc.	fem.
11	חדעסר		חדעסרא			
12	תרעסר			תרעסרא		<i>šinšeret</i>
13		תלתעסר		תלתעסרא		
14	ארבעסר	ארבעתעסר	ארבעסרא			
15	חמשעסר	חמשתעסר	חמשעסרא			<i>hamiššeret</i>
16		שתתעסר		שתתעסרא		
17	שבעסר	שבעתעסר	שבעסרא		<i>šebešer</i>	
18	תמנעסר	תמנתעסר	תמנעסרא		<i>šamanešer</i>	<i>šamanešeret</i>
19	תשעסר	תשתעסר	תשעסרא			

The disappearance of 'ayin in Akkadian could have influenced the development of the contracted teens not only in Akkadian,²⁰ but in Aramaic and Hebrew as well.

RECONSTRUCTION OF THE HEBREW CONTRACTED TEENS

Based on the two or three contracted teens in the Copper Scroll, as well on the appearance of contracted teens in various Aramaic dialects, it is possible to reconstruct the rest of the Hebrew contracted teens. The next table lists the proposed reconstructions (with the three cases in the Scroll underlined>). The possible spellings are with *samekh* and *šin*, as well as with 'alef and *he* endings.²¹

the teens (*Studies in Galilean Aramaic*, [1976], 26-28).

18 R.D. Wilson, *Elements of Syriac Grammar by an Inductive Method* (1891), 98.

19 E.g., אסרא והרא ירהא (= אסרא ירהא, "eleventh month," I 380:19), vs. תריסר (= תריסר רמו, "of the eleventh month," Z 20); תריסר דמו (= תריסר דמו, "twelve statues" (I 123:8 and I 95:14, respectively). See T. Nöldeke, *Mandäische Grammatik* (1875), 347-48.

20 Akkadian also has contracted teen ordinals: *ištenšerā*, "11th"; *šinšerā*, "12th"; *šalašerā*, "13th"; and *erbešerā*, "14th." See, S. Moscati, 117, 119.

21 It should be added that it is possible that the contracted teen became separated, or visa-versa. Perhaps it evolved in the following example indicates: -> תשע עשרה -> תשע עשרה, "nineteen." Thus, most suggestions in this table can be separated between the unit and the ten (before עשרה, עשרה, עשרה, or עשרה).

HEBREW CONTRACTED TEENS

#	masc.	masc.	fem.	fem.	fem.	fem.	notes
11	חדסר	חדשר	חדסרא	חדשרא	חדסרה	חדשרה	22
12	שניסר	שנישר	שתיסרא	שתישרא	שתיסרה	שתישרה	23
13	שלוסר	שלושר	שלוסרא	שלושרא	שלוסרה	שלושרה	24
14	ארבעסר	ארבעשר	ארבעסרא	ארבעשרא	ארבעסרה	ארבעשרה	25
15	חמסר	חמשר	חמסרא	חמשרא	חמסרה	חמשרה	26
16	שסר	ששר	שסרא	ששרא	שסרה	ששרה	27
17	שבעסר	שבעשר	שבעסרא	שבעשרא	שבעסרה	שבעשרה	28
18	שמונסר	שמונשר	שמונסרא	שמונשרא	שמונסרה	שמונשרה	29
19	תשעסר	תשעשר	תשעסרא	תשעשרא	תשעסרה	תשעשרה	30

WORD ORDER REGARDING NUMBERS AND NUMERALS

In the Scroll there are several patterns regarding the description of substances (e.g., silver, gold), units of measurements (e.g., talents, cubits), and numbers (e.g., ten, 21). These basic elements occur in various orders. Frequently one element is missing, and also there are a series of patterns. The patterns may be preceded by a verbal form (e.g., is buried or dig) or other defining words (e.g., a measurement), and six times they are followed by Greek let-

22 In the contracted forms the *'alef* and the *taw* would disappear. Cf. חר(י)סר(י) in Aramaic. חר (masc., A, once in BH, Ezek. 33:30) is the same as אחר - אחר (const. אחר). אחר (fem., pausal אחרת) may originate from אחרת or אחרת (cf. Ibn-Ezra and 'Avi-'Ezer on Gen. 11:6; J. Kimhi, *Sepher Sikkaron*, 27; Radak, *Mikhlol*, 185b). This can be compared with the Aramaic חרה (fem.) and חר (masc.). אחרת in Modern Hebrew means "single."

23 The Hebrew forms are שתי עשרה - שתיים עשר (masc.) and שתיים עשרה (fem.). Contracted teens would use forms without the *mem*. שתיים עשרה occurs in the Scroll. Cf. תרתיסר and תרסיסר (A). The Aramaic cognates of שני (masc.) - שני (const.) are תרתי (fem.) - שתי (const.) is תרתי (const.) - שתי (const.) is תרתי (const.) is "two (of)."

24 Cf. תליסר(י) (A) where the *taw* is assimilated into the *samekh*. The Aramaic cognate of שלושה (masc.) - שלושה (fem.) are תלת - תלת, "three" (*šin-taw* interchange).

25 Likewise in שבעעשרה one *'ayin* remained. Cf. ארבעיסר (TA) which may be the same as the Syriac ארבעסר, ארבעסר. In ארבי(י)סר(י) (A) both *'ayins* disappeared.

26 Probably the *šin* and *sins* of חמשה עשר (masc.) - חמש עשרה (fem.) would merge as in חמשה(י)סר(י) (CS) and חמשה(י)סר(י) (A).

27 Rather than ששסר(א) or שששר(א). The *'ayin* would disappear, the second *šin* would merge with the *šin*, as in שלושרא (CS). In שית(י)סר(י) (A) the *taw* did not assimilate. The Aramaic cognates of ששה (masc.) - ששה (fem.) are שש - שש, "six" (*šin-taw* interchange).

28 In שבעעשרה and תשעסר(י) one *'ayin* remained. Cf. ארבעיסר (TA) which may be the same as the Syriac ארבעסר(י). In the Aramaic ארבי(י)סר(י) both *'ayins* disappeared.

29 Cf. שלושרא (CS) and תמניסר(י) (A). The Aramaic cognate of שמונה (masc.) - שמונה (fem.) are תמני - תמני, "eight" (*šin-taw* interchange).

30 In תשעעשרה and שבעעשרה one *'ayin* remained. In the Aramaic תשעסר(י) both *'ayins* disappeared.

ters. Occasionally there are unusual word orders within a given phrase having these elements (e.g., 2:14-15; 3:12-13). In the subsequent discussion the numerals are treated as numbers. The abbreviation ככ, *kk*, like כְּכָרִין, "talents," is considered as substance. The Greek letters are disregarded.³¹

The following basic patterns are found in the Scroll. These are compared with biblical citations.

UNIT OF MEASUREMENT + NUMBER — אַמּוֹת אַחַת אַרְבָּעִין, "forty-one cubits" (1:3); אַמּוֹת שֵׁשׁ, "six cubits" (1:12); 42 כְּכָרִין, "42 talents" (2:2); כְּכָרִין תִּשְׁעַת מֵאוֹת, "nine hundred talents" (1:8); אִיסְתָּרִין אַרְבַּע, "four *stater*s" (9:3). Cf. אַמּוֹת עֶשְׂרִים, "twenty cubits" (2 Chron. 3:4,8,11,13).

SUBSTANCE + NUMBER — בְּדִין שֵׁל כֶּסֶף שֵׁשׁ, "six silver pitchers" (2:11; 7:10); וְכֵלִין כּוֹפְרִין עֶסְרָה וְעֶסְרִין, "plus ten/twenty sprinkling vessels" (10:9-11). Cf. כְּפֹרִי זָהָב שְׁלֹשִׁים, "thirty gold sprinkling vessels" (Ezra 1:10).

NUMBER + SUBSTANCE — שְׁנֵי דוּדֵינָא (const.), "two pots" (4:8); שְׁנֵי פְתָחִין (const.), "[the] two entrances" (6:1-2); שְׁתֵּי חֲפֹרֹת (abs.), "two pits" (9:2); בְּאַרְבַּע רוּחֹת, "with four sid[es]" (7:5); שִׁבְעַת בְּדִין, "seven pitchers" (9:2-3). Cf. אַרְבָּעָה כְּדִים מַיִם, "four pitchers of water" (1 Kgs 18:34).

SUBSTANCE + NUMBER + UNIT OF MEASUREMENT — כֹּסֶף אַרְבָּעִין [כ]כָּר, "forty [sil]ver [ta]lents" (1:14-15). Cf. זָהָב מֵאָה כְּכָר, "one hundred gold talents" (Ezra 8:26).

SUBSTANCE + UNIT OF MEASUREMENT + NUMBER — כֶּסֶף כְּכָרִין שִׁשִּׁין, "sixty-two silver talents" (10:6-7); כֶּסֶף ככ מֵאוֹת, "two hundred silver *kk*" (4:10). Cf. שֵׁשׁ מֵאוֹת, "six hundred silver talents" (Ezra 8:26).

UNIT OF MEASUREMENT/SUBSTANCE + SUBSTANCE + NUMBER — עֶשְׂתוֹת זָהָב, "sixty-five gold bars" (2:4); 100 עֶשְׂתוֹת זָהָב, "100 gold bars" (1:5-6). Cf. כַּפּוֹת זָהָב שְׁתֵּים עֶשְׂרֵה, "twelve gold pans" (Num. 7:84,86).

UNIT OF MEASUREMENT + NUMBER + SUBSTANCE — כְּכָרִין שְׁלוֹשׁ מֵאוֹת, "three hundred talents of gold" (10:11-11). E.g., לְשִׁקְלִים חֲמִשִּׁים זָהָב, "of fifty gold shekels" (2 Chron. 3:9).

VERBAL/NOMINAL FORM + UNIT OF MEASUREMENT + NUMBER — חָפַר אֶמָּה, "is buried (or dig) one cubit (deep)" (4:13-14); חָפַר חֲמֵשׁ אֶמָּה, "is buried (or dig) eight and half cubits (deep)" (9:5-6); מָשַׁח, "a distance of (or measure) from its bottom thirteen cubits" (9:1-2); מָשַׁח מֵאוֹת עֶסְרִין וְאַרְבָּעִין וְעֶשְׂרִין, "a distance of/-measure from its bottom twenty-[fo]ur/[se]ven cubits" (7:6-7); רָחוֹק מֵאוֹת, "a distance of nineteen cubits" (2:8).

A (TOTAL) (WEIGHT) OF + (UNIT OF MEASUREMENT) + NUMBER (+

31 Polzin discusses some of these orders, "Late Biblical Hebrew Toward an Historical Typology of Biblical Hebrew Prose" (1976), 60, 82 n. 79.

WEIGHT + NUMBER) — הַכֹּל מִשְׁקָל כִּכָּרִין 71 מִגִּין עֶסְרִין, “a total weight of 71 talents, twenty minas” (12:9); מִשְׁקָל כִּכָּרִין שִׁבְעָשָׂרָה, “a weight of seventeen talents” (1:4); כָּל שֵׁשׁ מֵאוֹת וְתֵשַׁע, “a total of six hundred and nine” (3:4).

UNUSUAL WORD ORDER — כָּה חֲפוּרֵי־חֲפוּר אֶמֶת בָּא אַרְבַּע for אַרְבַּע, “in it buried (or dig) four cubits (deep)” (2:14).

Finally, singular noun (without a definite article), with or without “one,” have identical meaning; e.g., אֶמְהָ = אַחַת, “one cubit” (4:14); שִׁדָּת כֶּסֶף = וְכִלֶּיהָ = שִׁדָּא אַחַת שֶׁל כֶּסֶף וְכִלֶּיהָ, “one silver carrying chair with its implements” (1:3), cf. וְכָל כִּלֶּיהָ = שִׁדָּא אַחַת וְכָל כִּלֶּיהָ, “one carrying chair with all its implements” (12:5); שָׁם קָלָל אֶחָד בּוֹ סֵפֶר אֶחָד = שָׁם קָלָל בּוֹ סֵפֶר אֶחָד, “three is one *qalal* (vessel), in it is one document.” On the other hand, a singular noun may be used as collective; e.g., [כָּ]סֶף אַרְבָּעִין [כָּ]כָר = כָּסֶף אַרְבָּעִין כִּכָּרִין, “forty talents of silver” (1:14-15; see Appendix A).

CONCLUSION

The Copper Scroll uses both written out numbers and numerical symbols. The pattern of the compound numbers generally follows that of Mishnaic Hebrew. The various word orders regarding numbers have parallels in (Late) Biblical Hebrew and in Mishnaic Hebrew. In the suggested reading of “forty-one cubits,” unit follows tens without the connective *waw*, which resembles the Phoenician - Punic usage. It is very interesting that the Scroll employs non-contracted and contracted teens. The latter is unknown from any other Hebrew sources, but can be found in various Aramaic dialects and in Akkadian. The three contracted Hebrew teens, plus the various Aramaic contracted teens, allow reconstruction of the rest of the Hebrew forms.

The fact that Aramaic non-contracted and contracted teens can be found in Megillat Ta'anit is very important because:

- (1) The Aramaic of Megillat Ta'anit proves that contracted teens were used during the Second Temple era.
- (2) Both the Aramaic Megillat Ta'anit and the Hebrew Copper Scroll use contracted and non-contracted teens.
- (3) Both Megillat Ta'anit and the Copper Scroll may originate from the Temple of Jerusalem and from the same era.
- (4) Both Megillat Ta'anit and the Copper Scroll are lists; the former of certain dates, the latter of hidden objects.

The Copper Scroll demonstrates that contracted teens were used at least in one Hebrew dialect. To this can be added the unusual formation of “forty-one.” These unique features, unknown from other Hebrew sources, distinguishes the language of the Copper Scroll from Biblical, General Dead Sea Scroll, and Mishnaic Hebrew.

APPENDIX C

THE MYSTERIOUS GREEK LETTERS

INTRODUCTION

There are seven sets of two-three Greek letters (some may be Latin) in the Copper Scroll. The following shows each line where these letters occur. The Greek letters are placed at the left end of the line for clarification.

LINES HAVING GREEK LETTERS

KEN	שקל ככרין שבועשרה	# 1, 1:4
XAT	אמות שש עד נקרת הטבילה	# 4, 1:12
HN	42 ככרין	# 6, 2:2
ΘE	השלשי (ל) עשתות זהב ששין וחמש	# 7, 2:4
ΔI	ובמוקא שבו ככרין עסר	# 9, 2:9
TP/TPI	40 ככ	#12, 3:7
ΣK	בצפוננו ככ]...[	#15, 4:2

Originally the Scroll was riveted together from three copper sheets, each containing four columns. The Greek letters are engraved only on the first sheet, and are always at the end of an item. They are distributed as follows: Column 1 has two sets of three letters; Column 2 has three sets of two letters; Column 3 has one set of two letters; and Column 4 has one set of two letters. Thrice the Greek letter-sets follow written out numbers, twice numerical symbols, once a lacuna where numerical symbols are missing, and once by a common noun.¹ There are about two spaces between שבועשרה and KEN, הטבילה and XAT,² עסר and ΔI, lacuna and ΣK; about four spaces between 42 and HN; about six spaces between 40 and TP;³ and no extra space between וחמש and ΘE.⁴ Since this line contains 22 Hebrew and two Greek letters, the engraver could not afford to leave out an extra space.

There is an additional *gamma* shaped letter at the beginning of the word

1 Some peculiarities of the Greek letters are noted by H. Bardtke, "Qumran und seine Probleme," *THR* NF 33, (1968), 193-94.

2 Milik (288) places a small circle for the first Greek letter. He is uncertain whether this letter is Σ, X, or Ζ.

3 Allegro (142 n. 69) says that an *iota* may follow TP, i.e., TPI, or it may be an extension of the shaft of the *lamed* below. McCarter reads "TP I 40" ("Mysterious," 40), but later without the *iota* ("Mysteryⁱⁱ," 236). The *iota* can be recognized in Baker's drawing, and perhaps in Milik's; it is sort of attachment to the upper vertical line of the *lamed* in Allegro's. The scholars read TP (see #12, 3:7).

האשוחין. Yet this is an unfinished *he*, i.e., האשוחין, rather than a *gamma* shaped 'alef, as some of the scholars propose (see #55, 11:12).

SUGGESTIONS BY THE SCHOLARS

ALLEGRO notes that the Greek letters in the Copper Scroll may "refer to the derivation of the sacred material hidden at these places." He compares them to the Hebrew and Greek code letters used to mark certain vessels mentioned in the Mishnah (m. Ma'as. Sh. 4:9-11; m. Sheq. 3:2).⁵ On the other hand, MILIK says that these Greek letters in a Hebrew document is inexplicable.⁶

LURIE proposes that the seven sets of Greek letters in the Scroll are codes for the seven branches of the main budget of Bar-Kokhba's Temple of Jerusalem (see "Major Publications on the Copper Scroll" in the Introduction). Lurie says that these are codes for certain funds, and cites Tannaic sources where codes marked on vessels are mentioned. Five times Lurie in his translation inserts before the Greek letters the phrase in parenthesis, "(and they are from the fund whose code is)" (1:4, 1:12, 2:4, 2:9, 3:7).⁷

ULLENDORFF says that the "connexion between the numerical values of these letters and their context is too striking to invoke mere coincidence." He suggests the following solutions:

- XΛΓ represent χίλια three = 3000, but it cannot be checked since there are no figures. 3000 may be an inflated figure of 300, as 170 for 17, and 400 for 40.
- KEN following שבעשרה, "seventeen," is 170 (i.e., 20 + ἑχατόν + 50 = 170); ten to one ratio between 17 and 170, while 170 may be an inflated figure.
- ΔΙ following עשר, "ten," can be explained as I = 10, and Δ is an abbreviation for Δέκα, "ten."
- ΞΕ following ששין וחמש, "sixty five," is 65. The first letter is badly written, but ΞΕ is better than ΘΕ.
- ΗΝ following ככרין, "42 talents," is 42. N = 50 and H = 8, since H is placed on the left, it is subtracted, 50-8 = 42; cf. the Roman numeral system.
- TP following ככ, "40 kk," is equal 300 + 100 = 400. There is a ten to one ratio between 40 and 400, while 400 may be an inflated figure.
- ΣΚ following [...] ככ cannot be solved due to the lacuna.⁸

Indeed, ΞΕ is equal 65, although according to the drawings the reading is ΘΕ. However, with the other solutions there are problems. E.g., the phrase "ΔΙ עשר" would indicate number "ten" three times.

4 Ullendorff reads ΞΕ, as discussed later.

5 Allegro, 136 n. 9. Allegro^{PF} places periods at the end of lines where the Greek letters are, except at the end of line 2:2. He also recognized the *gamma* in line 11:12.

6 Milik, 221 A-5; 285; Milik^E, 144.

7 In the other two cases (2:2, 4:2) there are no translation for the items (presumably they were inadvertently omitted). Lurie, 56-58 and n. 8, 65, 67, 69, 70, 76, 79. (On p. 56 line 9, "Milik" instead of "Allegro" is also a typographical error.)

LEHMANN suggests that XAT (1:12) is like the Hebrew חג ("pilgrimage, holiday").⁹ Similarly, one can suggest the possibility that TP (3:7) may be the initial of the word TPTMA, *terumah* (תרומה, "elevated gift, priestly share"), a term not used in the Copper Scroll.¹⁰

PIXNER suggests that the Greek letter sets are the initial two or three letters of contemporary Greek personal names, who might have been responsible for certain treasures. Such names can be found in Josephus' writings, yet Pixner claims no relationship between the two, except ΘΕ for ΘΕΒΟΥΤΙΣ or ΘΕΒΟΥΘΙ (2:4), which he seeks to identify with the Priest Thebouthi son of Jesus (War vi, 8, 3). Pixner connects Thebouthi with an alternate translation of his reading of בית המדח ישו, "the banished (cast-off) House of Jesus" (2:3-4). This plus his suggestion concerning the name of Matthia in the following item (2:5), indicate that Pixner deliberately attempts to 'Christianize' the Copper Scroll (see #7, 2:3-4; #8, 2:5).¹¹

BEYER says that the Greek two-three letters are pronounceable syllables. Like Pixner, he suggests that the Greek letters are the beginning of names or word, rather than numerals. Beyer endorses three names from Pixner's suggestions, and for four he offers his own. The additional letters he places in brackets. The chart below shows both Pixner's and Beyer's suggestions.

	PIXNER ¹²			BEYER ¹³	
(1:4)	KEN	KENEΔΑΙΟΣ	Kenedaios	Κενεδαῖος	Kenedaios
(1:12)	XAT	XATEIPAS	Chageiras	Χαγειρας	Chageiras
(2:2)	HN	HNNΑΦΗΝ	Ennaphen	Ηνίοχος	Eniochos
(2:4)	ΘΕ	ΘΕΒΟΥΤΙΣ Θεβουθί	Theboutis Thebouthi	Θεόδοτος	Theodotos
(2:9)	ΔΙ	ΔΙΟΦΑΝΤΟΣ ΔΙΟΓΕΝΕΣ	Diophantos Diogenes	Διονύσιος	Dionysios
(3:7)	TP	TPTΦΟΝ	Tryphon	Τρύφων	Tryphon
(4:2)	ΣΚ	ΣΚΟΠΑΣ	Skopas	Σκενᾶς	Skeyas

STEGEMANN accepts that the Greek letters are the beginning of personal names. However, he believes, these names refer to people from Adiabene, rather than to the Essenes as Pixner suggests. The Adiabene family had palaces in Jerusalem, were involved with the fight with the Romans, and some survived the war. The money may have been theirs and were given to the priests for safekeeping. EISENMAN agrees with Stegemann.¹⁴

8 "The Greek Letters of the Copper Scroll," VT 11 (1961), 227-28.

9 "Identification," 105 n. 6.

10 תרומה is a synonym of דמיע; see Appendix D.

11 See the next footnote.

12 Regarding the names, Pixner refers to War i, 51, 113, 529, 547; ii, 520; v, 474; vi, 384; Ant xii, 3, 3, 132. Pixner, 335-36 and n. 32, 344-46 and nn. 11-12.

13 Beyer, 224-27.

MCCARTER notes that the the limited number of Greek letters do not correspond to any known system of abbreviation. Any suggestion may be correct, but it is difficult to justify any conclusion.¹⁵

GORANSON says that the Greek letters may relate to removal or withdrawal of deposits. Perhaps these items were discovered by Romans or non-Essenes, and the Greek letters would indicate canceling of these deposits.¹⁶

THIERING proposes that the Greek letters represent numerals, but refer to deposits of separately kept Greek coins. The most important (gold) is placed in the center, the next (silver) on the right, and the next (copper) on the left. Thus, *chi-alpha-gamma* would mean 600 copper, 1 gold, and 3 silver coins; *tau-rho* would be 300 copper, 100 silver, with no gold coins.¹⁷ This is a highly speculative hypothesis. Why is gold at the center, silver to its right, and copper to its left? Why not a different order?

3Q15 VERSUS 4Q186

Random use of Greek letters in a Hebrew document may be a reflection on its cryptic nature. The horoscope 4Q186 is written with Hebrew square characters from left to right. Some words and letters were written with paleo-Hebrew, Greek, and cryptic scripts of unknown origin.¹⁸ Since this document is a horoscope, perhaps it was purposely made difficult to read, and it was used for mystical purposes. Somehow 4Q186 is similar to several so-called “mirror written text” within the Ugaritic corpus, written from right to left, like in Hebrew, instead of the Ugaritic system from left to right.¹⁹

Although both 3Q15 and 4Q186 are written in Hebrew and both contain Greek letters, one cannot connect the two documents for the following reasons: (a) The Copper Scroll is a list of hidden treasures where the Greek letters may indicate the sources or the controllers of the particular funds. (b) In the Copper Scroll the Greek letters cannot be read as meaningful words; they seem to be abbreviations. (c) 4Q186 is a horoscope written in the opposite direction. (d) In 4Q186 the Greek, paleo-Hebrew, and cryptic letters (except one) can be read as meaningful words within their respective phrases. Nevertheless, 3Q15, as well as 4Q186 indicate that for one reason or another, it might have been a practice to use Greek letters in Hebrew documents.

14 See next note.

15 “The Copper Scroll Treasure,” 147-48; “Mysterious,” 40; “Mysteryⁱⁱ,” 236-37.

16 “Further Reflections on the Copper Scroll,” 6-7.

17 B.E. Thiering, “The Copper Scroll — King Herod’s Bank Account?” Paper presented at the International Symposium on the Copper Scroll, Manchester, September 9, 1996.

18 *DJD V*, 88-91, pl. xxxi. Allegro suggests that the word תוארינו (3:3) which in mirror reading equals מיראות may be a scribal error for מילאות (p. 90). However, this simply may be a *lamed-reš* letter interchange.

19 *UT*, 16.

NUMERICAL VALUES OF THE GREEK LETTERS

The Greek letters have numerical values, as do Hebrew and Arabic. In these systems usually first the larger unit is written, then the smaller one. Yet both orders are attested.²⁰ These are different from the Roman numerals, where only a few letters are used, and when the smaller unit is placed to the left of the larger one, it is subtracted. The chart below shows the stated substances and values, the Greek letter sets, the numerical value of the Greek letters, and the expected Greek forms for these values.

	STATED SUBSTANCES	STATED VALUES	GREEK LETTERS	GREEK VALUES ²¹	EXPECTED FORMS
(1:4)	talents	seventeen	KEN	75	EO or OE
(1:12)	---	---	XΑΓ	604	ΔX or XΔ
(2:2)	talents	42	HN	58	HN or NH
(2:4)	gold bars	sixty-five	ΘΕ	14	ΔΙ or ΙΔ
(2:9)	talents	ten	ΔΙ	14	ΔΙ or ΙΔ
(3:7)	kk	40	ΤΡ	400	Τ
(4:2)	kk	[...]	ΣΚ	26	ΣΚ or ΚΣ

As can be seen, only three out of seven sets would conform with the Greek system. However, none of these matches with the stated values. XΑΓ does not have any possible corresponding value, while ΣΚ is followed by a lacuna. If these values would be identical with those stated, it would indicate that they are repeated, as often in Phoenician inscriptions (e.g., *KAI* 1, 3 [#14:1]; see Appendix A), in the Elephantine Documents (e.g., Kraeling 12:5-5; see #38, 9:3), etc. But here the case is not so. Thus, one can ask: What would these values indicate after an already given number of the substances? If the Greek letters represent cryptic numbers, why would one indicate the number or numeral value first in a non-secret manner? If the compilers possessed or developed a secret numerical code, why was not this code employed exclusively? Therefore, it is unlikely that the Greek letters in the Scroll represent numerals.

Furthermore, in the Hebrew coins from the time of the Jewish - Roman War, often the year is indicated by the numerical value of the 'alef-bet. E.g., שקל ישראל א ירושלם קדשה, "Shekel of Israel (year) 1 Jerusalem the Holy," where א means "(year) 1" of the war (= 66/67 CE); שקל ישראל שב ירושלם, "Shekel of Israel y(ear) 2 Jerusalem the Holy," where שב is an abbreviation for ב שנת, "y(ear) 2" (= 67/68 CE); and so for שג, "y(ear) 3"

20 E.g., year 14 (= 74 CE) among the Agrippa II coins can be found both ways, ΙΔ and ΔΙ. See also footnote 22.

21 The numerical values of the Greek letter sets are: KEN = 20+5+50; XΑΓ = 600+1+3; HN = 8+50; ΘΕ = 9+5; ΔΙ = 4+10; ΤΡ = 300+100; ΣΚ = 6+20.

(= 68/69); Ψ , “y(ear) 4” (= 69/70); Σ , “y(ear) 5” (= 70/71).²²

Seemingly the Copper Scroll was compiled just before the Jewish War started. Although it is unknown when the alphabetic numerical system was introduced in Hebrew, it had to be around for a long time.²³ It is interesting that the compilers of the Scroll employed the old numerical symbols, rather than the alphabet. Yet it is possible that the old symbols were selected because they might have been known only to a few people, and thus they served to camouflage some of the listed values in the Scroll.

ARE THE GREEK LETTERS PART OF THE DOCUMENT?

One can argue that the Greek letters have nothing to do with the Hebrew text. Perhaps the first sheet was prepared for a Greek text and these letters were part of it, but later it was used for a Hebrew inscription. Therefore, the Greek letters appear always at the left side of the text, since Greek is written left to right. For the engraver it would be difficult to remove these letters. Maybe the compilers did not care about them, the same way as some scribal errors could be corrected, but were not; e.g., the Γ into a η (9:11).

However, the Greek letters are always in line with the Hebrew text, and always at the end of an item. Some of the letters resemble Hebrew characters; e.g., the P of TP (3:7) has almost identical shape with the *qofs* of ... מִזְרְקוֹת (3:3-4); the H of HN (2:2) resembles the shapes of *he* and *het* in the same column. Thus, probably the same hand engraved the Greek and Hebrew letters of that line where both appear. In addition, the Greek letters never interfere with the Hebrew text, and there is enough space left out between the Hebrew and Greek letters, except in line 2:4. This may also indicate that first the Hebrew characters were engraved, and after that was added the Greek letters. Therefore, the Greek letters are an integral part of this unusual document.

SUGGESTED INTERPRETATIONS

The following interpretations of the Greek letters are a combination of the various suggestions offered by Allegro, Lurie, Pixner, and Beyer. Although far from being the proper solution, presently it appears to be the best one.

22 For the Greek and Hebrew coins, see A. Reifenberg, “Ancient Jewish Coins,” *JPOC*, 19 (1941), 296-98, 302-03; A. Kindler, “Coins and Currency,” *EJ* 5, 699-700, 706-07, 719.

23 The scholars suggest that it was adapted from Greek, yet some of the *gematriot* (pl. of *gematria*, “numerological method,” from $\gamma\epsilon\omega\mu\epsilon\tau\rho\acute{\iota}\alpha$) used in the Talmud - Midrashic literature may go back to the time of Jeremiah (cf. Baraita of the Thirty-Two Rules by Rabbi Eliazar son of Rabbi Yose hagGelili, Rule 29). See also, G.R. Driver, “The Abbreviations in the Masoretic Text,” *Textus* 1, 114, 126-27 (although this writer disagrees with many of his suggestions).

SUGGESTED SOLUTIONS

(1:4)	(Code:)	KEN	=	KENEΔΑΙΟΣ	Kenedaios	
(1:12)	(Code:)	XAG	=	ΧΑΓΕΙΡΑΣ	Chageiras	
(2:2)	(Code:)	HN	=	ΗΝΝΑΦΗΝ	Ennaphen	or
				ΗΝΙΟΧΟΣ	Eniochos	
(2:4)	(Code:)	ΘΕ	=	ΘΕΒΟΥΤΙΣ	Theboutis	or
				ΘΕΒΟΥΘΙ	Thebouthi	or
				ΘΕΟΔΟΤΟΣ	Theodotos	
(2:9)	(Code:)	ΔΙ	=	ΔΙΟΦΑΝΤΟΣ	Diophantos	or
				ΔΙΟΓΕΝΕΣ	Diogenes	or
				ΔΙΟΝΥΣΙΟΣ	Dionysios	or
(3:7)	(Code:)	ΤΡ	=	ΤΡΥΦΟΝ	Tryphon	
(4:20)	(Code:)	ΣΚ	=	ΣΚΟΠΑΣ	Skopas	or
				ΣΚΕΤΑΣ	Skeyas	

"Code" indicates that the Greek letters are used as such. Obviously many more contemporary Greek names can be found.²⁴

CONCLUSION

The Copper Scroll is a list of hidden silver, gold, and vessels, etc., probably from the Jerusalem Temple. Since five times the Greek letters follow stated numbers or numerals, plus the Greek ciphers do not match them, the Greek letter sets cannot represent numerical values. Maybe they are initials of Greek common names, as Pixner and Beyer suggest. The letter sets may indicate that some of these objects originated or were controlled by Greek speaking Jews, also that the compiler(s) were likely familiar with the Greek language.

At the very end of the Scroll it is mentioned where a detailed copy of the document can be found. Perhaps this document could clarify the correct meaning of the Greek letter sets. Yet it is doubtful whether that copy will ever be uncovered. Thus, the significance and mystery of the Greek letters in the Scroll may remain unsolved.

24 Chageiras son of Nabateus of Adiabene was named so because he was unfortunately lame (War, V xi 5). Thus, the name is derived from the Aramaic חֲגִירָא - חֲגִירָא or Hebrew חֲגִיר, "lame." The Targum renders פֶּרְעֵה נְכֹאֲנִכוּ מֶלֶךְ מִצְרַיִם, "Pharao Necho king of Egypt," as פֶּרְעֵה חֲגִירָא מֶלֶךְא דְּמִצְרַיִם, "the lame Pharao the king of Egypt" (2 Kgs 23:29; Jer. 46:2). The name נְכֹאֲנִכוּ is like the phrase נִכְה רַגְלִים, "lame on the feet" (2 Sam. 4:4), and this is the understanding of the Targum (see Radak on the first, and *Mešudat Zion* on both verses; see also Rid and *Mešudat Zion* on 2 Sam. 4:4).

APPENDIX D

THE TERM כלי דמע

INTRODUCTION

דמע is the most important term in the Copper Scroll. Thus this lengthy appendix is devoted to a multifaceted discussion of this word.¹ דמע occurs in ten items for a total of fourteen cases. In five items the *dema*^c vessels are followed by the problematic phrase בְּתִכּוֹן־בְּתִכּוֹן אֶצְלָמוֹן (see Appendix E).

כלי דמע	בלגין ואפורין	הכל של הדמע ...	# 4
כלי כסף וזהב של דמע	מזרקות כוסות מנקיאות קסאות ...		#12
כלי דמע	ולבושין ...		#13
כאלין של דמע	ובתכן \ וכתבן	אצלמ	#22
כלי דמע	וספרין אל הכלין		#33
בכלי דמע	בתכן \ כתבן	אצלמ	#50
כלי דמע	דמע סנה ותכן \ ותבן	אצלמ	#51
כלי דמע	דמע או דמע סות	ובתכן \ וכתבן	#54
כלי דמע <ע> לאח	דמע סירא	בתכן \ כתבן	#55
כל <י> כסף וכלי זהב של דמע	וכסף ...		#58

The fourteen cases of דמע can be classified in seven groups as follows:

- (1) כְּלִי־כְּלִי שֶׁל דְּמַע = כְּלִי־כְּלִי דְּמַע (#4, 1:9; #13, 3:9; #33, 8:3).
- (2) בְּכִלִּין־כְּכִלִּין שֶׁל דְּמַע = בְּכִלִּין־כְּכִלִּי דְּמַע (#50, 11:1).
- (3) כְּלִי־כְּלִין שֶׁל דְּמַע (שֶׁל) ... וְכִלִּין־כְּלִין שֶׁל דְּמַע שֶׁל ... = כְּלִי־כְּלִי דְּמַע ... דְּמַע ... (#51, 11:4; #54, 11:10; #55, 11:14).
- (4) כְּלִי דְּמַע = כְּלִין שֶׁל דְּמַע = כְּאֵלִין שֶׁל דְּמַע (#22, 5:6-7).
- (5) כְּלִין ... שֶׁל דְּמַע וְכִלִּין ... שֶׁל דְּמַע = כְּלִי <י> ... וְכִלִּי ... שֶׁל דְּמַע (#58, 12:6-7).
- (6) כְּלִין שֶׁל ... שֶׁל דְּמַע וְכִלִּין שֶׁל ... שֶׁל דְּמַע שֶׁל = כְּלִי ... וְכִלִּי ... שֶׁל דְּמַע (#12, 3:2-3).
- (7) הַכֵּל <י>־הַכֵּל <י> שֶׁל הַדְּמַע־הַדְּמַעוֹת or even הַכֵּל שֶׁל הַדְּמַע־הַדְּמַעוֹת (#4, 1:10).²

THE TERM כלי

The term כלי (pausal כְּלִי, const. כְּלִי, pl. כְּלִיםִין, const. כְּלִי, BH, MH) has several meanings, such as “vessel, object, implement, tool, garment, cover.” It occurs 16 to 19 times in the Scroll.³ The vocalization of כלי depends on

1 This appendix represents what would be parts of ten individual items, as well as an in-depth analysis of the subject.

2 Although the latter is unlikely, it is a possible reading. Accordingly דְּמַעוֹת would be the plural of דְּמַע (see #4, 1:10).

3 כלי (1:9, 3:2, 3:9, 8:3, 11:4, 11:10, 11:14); בכלי (11:1); וכלי (12:6); כלין (2:6, 2:8);

the content: כְּלִי, "a vessel," כְּלִי שֶׁל = כְּלִי, "a vessel of," or כְּלִי = כְּלִין שֶׁל, "vessels of." Also, כְּלִי or כְּלִי may be a shortened plural form instead of כְּלִים, "vessels" (see the discussion of כְּלִין later). Twice כָּל can be vocalized as or amended to כָּל־וְכָל, "vessel(s) of" (see #58, 12:6), and once הַכָּל as הַכָּל - הַכָּל, "the vessels" (see #4, 1:10). Once the *plene* spelling of כְּלִין is used instead of כְּלִין (#22, 5:6-7).

כְּלִי and כְּלִי are often further defined; e.g., כְּלִי חֶרֶשׁ, "earthen vessel" (Lev. 6:21, etc.); כְּלִי עֵץ, "wooden vessel" (Lev. 11:32, etc.); כְּלִי שֵׁרֶת, "service vessel" (m. Sot. 2:1); וְשָׁם כְּלִי הַמִּקְדָּשׁ וְהַכֹּהֲנִים, "and there are vessels of the Temple and the priests" (Neh. 10:40); כְּלִי כֶסֶף וְכִלְיָיִם, "silver objects and gold objects" (Gen. 24:53; Exod. 3:22); כְּלִי לָבָן, "white garments" (m. Ta'an. 4:8); also,

וְכָל כְּלִי מִשְׁקָה הַמֵּלֶךְ שְׁלֹמֹה זָהָב (And all the drinking vessels of King Solomon were of gold), וְכָל כְּלִי בֵּית יַעֲרֵן הַלְבָּנוֹן זָהָב טָהוֹר (and all the vessels of the House of the Forest of Lebanon were of pure gold) (1 Kgs 10:21).

THE SCHOLARS INTERPRETATIONS OF כְּלִי (כלי)

The term כְּלִי (כלי) has been analyzed as follows. ALLEGRO^{pr}: דָּמַע, "juice" of wine, "oil"; cf. Exod. 22:28. כְּלִי דָּמַע, "vessels of/for *dm^c* oil" (1:9, 3:2-3, 3:9, 5:7, 8:3); ... כְּלִי דָּמַע אוֹ דָּמַע, "vessels of *Dema^c* oil or ... oil" (11:10); *Dema^c*, since the vessel is not defined; "vessels of oil" (11:1); "a vessel (vessels) of ... oil" (11:4); "a vessel filled with oil of" (i.e., כְּלִי (ד) "כלי מלאה דָּמַע" (11:14); הַכָּל שֶׁל דָּמַע, "whole being of *dm^c* oil" (1:10). The text has כְּלִי דָּמַע, i.e., the third letter, a *yod*, looks like a *dalet*, a dittograph or an error for *yod* (i.e., כְּלִי (ד) דָּמַע, or כְּלִי דָּמַע). For כְּלִי without a *yod*, cf. 12:6. Also, כְּלִיד is a "cup" in Aramaic (Jastrow, 642a); i.e., כְּלִיד דָּמַע, "a cup of *dm^c*" (11:4), and כְּלִיד דָּמַע, "a cup filled with oil of" (11:14).⁴ However, this was changed to, "vessel of ... oil, oil of" (i.e., כְּלִי דָּמַע ... דָּמַע). In כְּלִי מִלָּאָה ... בָּהּ the gender of כְּלִי is questionable. And changed again to "vessel of liquid oil" (i.e., כְּלִי דָּמַע לָאָה). ALLEGRO^{pr}: כְּלִי דָּמַע are "vessels of/for tithe, tithe vessels/jars," i.e., vessels containing or that once contained tithe. Vessels which held tithe were no longer usable for secular purposes had to be hidden.⁵ MILIK: כְּלִי דָּמַע are "vases containing aromatic spices." דָּמַע is from the root דָּמַע, "to shed tears," i.e., tears of trees, thus spices. דָּמַע in וְדָמַע is interpreted by the Rabbis as a synonym of תְּרוּמָה. Milik cites a fragment from 4Q Halaka^a 5 which contains the term דָּמַע, "prémices du blé," or "first fruits of wheat." Yet Milik maintains that דָּמַע in the Scroll means "aromatic spices."⁶

כְּלִין (5:6); כְּלִין (8:3); וְכִלְיָיִם (10:11); כְּלִי (12:5); וְכִלְיָיִם (1:3); and כְּלִי (12:6, 12:8), and כְּלִי (1:10, alt.).

4 This correction was made by Allegro on or prior December 1, 1955.

5 Allegro, 37, 41, 47, 53, 55, 137 n. 19, 141 n. 62; Allegro^{rev}, 21, 22, 24, 26, 27.

DUPONT-SOMMER: כלי דמע are “vases containing aromatic spices” (Milik), rather than “vessels containing tithe” (Allegro).⁷ GEVARYAHU: כלי דמע are various types of vessels prepared to store תרומה and קודש (consecrated objects), and mentions the relevant Talmudical sources.⁸ LURIE concurs with Gevaryahu. דמע, based on Rabbinical sources, is the same as תרומה, “priestly share.” כלי דמע means “vessels of priestly share.” Once Lurie renders כלי ... של תרומה as כלי ... של דמע (3:2-3), i.e., “... vessels of priestly share,” while for the other cases כלים של דמע, although in Modern-Hebrew it is an unknown term. כלי דמע is a sort of vessels; one of three types in Item 4, and one of five types in Item 12.⁹ LEHMANN points out that Allegro’s view contradicts the Mishnaic Halakhah that a jar emptied from liquid *terumah* may be used for non-sacred purposes. He also rejects Milik’s translation. דמע is a synonym of תרומה, “heave offering.” Its secondary meaning is an admixture of תרומה (“sacred”) and חולין (“non-sacred”) produce. כלי דמע are “heave-offering vessels,” i.e., vessels holding heave-offerings, rather than “tithe vessels.”¹⁰ SHARVIT discusses the term ודמעך (BH), as well as the Mishnaic usage of דמע. The meaning of כלי דמע is doubtful, since the companion words are not yet understood.¹¹ Apparently, these are סנה, סירא, סוח, לאח. GREENFIELD refutes Milik’s interpretation. דמע in the Bible means “the best, fat part,” and is applied to cereals, while in the Mishnah it is equivalent to תרומה. Thus, כלי דמע refers “to a particular type of jar used for storing תרומה and other types of sacred gifts.” Greenfield notes the Samaritan דמע, Arabic *dimaḡ*, and says that the two roots of דמע must be kept apart.¹² SARFATI: כלי דמע were filled with fruits. The reading is not כתבן אצלסון, rather בתכן אצלסון, “their documents are with them,” which shows whether the content is תרומה (“priestly share”), מעשר (“tithe”), or בכורים (“first fruit”).¹³ QIMRON agrees with Sarfati.¹⁴ PIXNER: דמע, “tithe”; כלי דמע, “tithe vessels,” “vessels of tithe,” or “vessels of tithing.” דמע refers to the priestly dues (Jastrow, 314). Pixner notes that Milik “prefers another translation which fits his theory,” but in his

6 Milik, 268, etc., 250 C-122, 300 “Addenda”; Milik^E, 139, etc.

7 A. Dupont-Sommer, “על מגילות-הנחושת 3,15 מקומראן” in *Report - Third World Congress of Jewish Studies* (1961), 92-93.

8 H.M.Y. Gevaryahu, לאור מקורות שבמגילת הנחושת 3,15 מקומראן “הערות לרשימת המסומנים שבמגילת הנחושת 3,15 מקומראן” in *Report - Third World Congress of Jewish Studies* (1961), 94-95 (there may be a typographical error, perhaps it should read, “כלי דמע means various types of vessels ...”).

9 Lurie, 62-63 n. 2-3, 74-75 and n. 5, 76, 79, 86-87, 98, 117-22, 124-25.

10 “Identification,” 97-101.

11 “Iyyunim,” 131-32 and n. 37א; “He‘arah le‘Iyyunim,” 150.

12 Greenfield, 139.

13 “Hiddah,” 107, 110-11, and English summary, np.

14 “He‘arah,” 518.

Addenda he agrees that it may mean "priestly share."¹⁵ YADIN: כלי דמע, "Dema^c (?) utensils." He compares the *dema^c* vessels of Item 12 with the altar utensils listed in the Temple Scroll, and with the utensils of the altar and table of show-bread in the Masoretic Text.¹⁶ WILMOT: דמע, "votive object"; כלי דמע, "votive vessels." דמע is a synonym of תרומה.¹⁷ WOLTERS: דמע "tribute"; כלי דמע, "vessels of tribute."¹⁸ GARCIA MARTINEZ: "tithe"; "tithe vessels" and "vessels of tithe."¹⁹ BEYER: דמע, "priestly share"; כלי דמע, "vessels with priestly share"; כאלין של דמע, "boxes (αἱ χηλοί) with priestly share."²⁰ MCCARTER: "vessels of *dema^c*"; "*dema^c* refers to produce that has been set apart (*herim*) as a special portion for the support of the Levites (*teruma*)."²¹ VERMES: "offering"; "vessels of offering."²² WISE (follows Wilmot): "votive vessels, vessels of votive offering."²³

THE MEANING OF דמע

Verbal forms of the root דמע, "to weep" (*Qal* inf. abs. and imperf. third pers. sg.) appear only in the verse, וְדָמַע תְּדָמַע וְתָרַד עֵינַי דְּמָעָה (Trg. וּמְדָמַע וּמְדָמַע, "and weeping it shall weep and shall run down my eye tears" (Jer. 13:17). This also contains דְּמָעָה (fem.), "tears." Verbal forms of דמע, as well as דְּמָעָה and דָּמַע (masc.), "tears," are common in the Talmudim, Midrashim, and Targumim.²⁴ דמע in the Scroll may be in the ab-

15 Pixner, 343 and nn. 7-8, 346-48, 350, 355-57.

16 *The Temple Scroll* (Heb.) 1, 174-78; (Eng.) 1, 227-29. Yadin's chart includes only the vessels listed in Item 12.

17 "The Copper Scroll of Qumran (3Q15) and the Greco-Roman Temple Inventories," 13-14; "David J. Wilmot and the Copper Scroll," 17-18.

18 "Notes," 590-91, 593-94; Wolters, 32-33, 36-37, 40-41, 46-47, 52-55.

19 Garcia Martinez, 461-63.

20 Beyer, 226-27, 229, 231-32.

21 "The Copper Scroll Treasure," 134, etc.

22 Vermes, 374-78 ("containers" in line 1:9 is apparently a typographical error).

23 Wise, 191-93, 195, 197-98.

24 Regarding דמע in the Targumim, cf. H.J. Kasovsky, *Ozar Leshon Targum Onkelos*, 136a; E.G. Clarke, *Targum Pseudo-Jonathan of the Pentateuch, Text and Concordance* (1984), 166a, 694 c. 5. *Segolite* nouns of the penultimate pattern פָּעֵל (*Pe'el*; usually masculine; pl. פְּעָלִים, *Pe'alim*/n, and פְּעָלוֹת, *Pe'alot*) are very common. When the third radical is a *het* or *ayin*, the second *segol* changes into a *patah*. The counterpart of *Pe'el* (e.g., דָּמַע, masc.) is פְּלָעָה (*Pil'ah*, usually feminine; e.g., דְּמָעָה, fem.), its plural pattern is פְּעָלוֹת (*Pe'alot*; e.g., דְּמָעוֹת). There are a number of *Pe'el* formations in the Scroll, e.g., אֶבֶן (fem.), "stone" (8:5, 10:9, 12:2; pl. אֲבָנִים); possible יֶגֶר (prob. masc., 4:13, 6:14, 8:8), "heap" (pl. יִגְרִים, masc.), "vineyard" (10:5; pl. כְּרָמִים); כֶּסֶף (masc.), "silver" (1:3, 2:11, etc.; pl. כְּסָפִים); נֶפֶשׁ (fem.), "tomb" (1:5; pl. נִפְשׁוֹת); עֶשֶׂת (masc.), "bar, mass" (pl. עֲשֻׁתוֹת, 1:5, 2:4); קֶבֶר (masc.), "grave" (3:11, 5:12, etc.; pl. קְבָרִים, 12:11); פֶּתַח (masc.), "opening, entrance" (1:8, 10:10, etc.; pl. פְּתָחוֹת, 6:2); סֶלַע (masc.), "rock" (2:11, 11:5; pl. סְלָעִים); דְּמָעוֹת the plural of דָּמַע, appears as an alternate reading (1:10). דָּמַע and דְּמָעָה, "tears," are of those nouns where the plural is formed in Hebrew with the suffix וֹת (i.e.,

solute, i.e., “*dema*’,” or in the construct state (i.e., *dema*’ ... of.”

In the verse *מְלֶאֱהָ וְדִמְעָה לֹא תִאָּחֵר*, “You shall not delay your *mele’ah* and your *dema*’” (Exod. 22:28),²⁵ *דִּמְעָה* means a sort of *תרומה*, “priestly share,” as the Rabbies explain:

מְלֶאֱהָ (your [sg.] *mele’ah*), these are *בְּכוֹרִים* (first fruit offerings) that are taken from the *מְלֶאֱהָ וּמְלֶאֱהָ* (*mele’ah*), and *דִּמְעָה* (your [sg.] *dema*’), this is *תרומה* (*terumah*) which is called *דִּמְעָה* (*dema*’; Mekhilta’ deRabbi Simeon ben Yohai on Exod. 22:28; cf. Yalqut Šim’oni 111).²⁶

מְלֶאֱהָ (your [sg.] *mele’ah*), these are *בְּכוֹרִים* (first fruit offerings) taken from the *מְלֶאֱהָ וּמְלֶאֱהָ* (*mele’ah*), and *דִּמְעָה* (your [sg.] *dema*’), this is *תרומה* (*terumah*; Mekhilta’ deRabbi Ishmael on Exod. 22:28).²⁷

מְלֶאֱהָ (your [sg.] *mele’ah*), these are *בְּכוֹרִים* (first fruit offerings), and *דִּמְעָה* (your [sg.] *dema*’), these are your *תרומה* (*terumah*; b. Tem. 4a).²⁸

(God) created (all) lands, He took the Land of Israel as a *תרומה* from all the lands; as it is written, “for the Lord is the land and *מְלֶאֱהָ* (its fullness)” (Ps. 24:1), and *מְלֶאֱהָ* does not mean anything but *תרומה*, as it is written, “*מְלֶאֱהָ וְדִמְעָה* (your *mele’ah* and your *dema*’; Exod. 22:28)” (Midrash Ps. on 24:1).

Likewise, the Targumim render *מְלֶאֱהָ וְדִמְעָה* as *בְּכוֹרֶךָ וְדִמְעָךָ*, “your first fruit offering and your *dema*’” (TO); *בִּיכּוּרֵי פִירְךָ וּבִיכּוּרֵי חֲמֶר נַעְוֹךָ*, “your first fruits offering, and first wine offering of your press” (PsJ); *מַעֲשֵׂרְתְּכוֹן וְדִמְעָכוֹן*, “your (pl.) tithes and your (pl.) *dema*’s” (TN);²⁹ *מְלִיתֶךָ וְדִמְעָךָ* (= *מְלִיתֶךָ וְדִמְעָךָ*), “your *mele’ah* and your *dema*’” (STA, STJ).³⁰

Also, the Targum of *קֹדֶשׁ יִשְׂרָאֵל לַיהוָה רִאשִׁית תְּבוּאָתָהּ*, “Israel is sanctified for the Lord, the first-fruit of His produce,” *קֹדֶשׁ יִשְׂרָאֵל אֵינָנוּ בֵּית יִשְׂרָאֵל קֹדֶשׁ* is *קֹדֶשׁ יִשְׂרָאֵל אֵינָנוּ בֵּית יִשְׂרָאֵל קֹדֶשׁ* ... *כְּדִמְעָה אֲרָמוֹת עַל־לֹא* ... *ה’*, “The House of Israel are sanctified before the Lord, ... like the *dema*’ harvest offering” (Jer. 2:3). Thus, *קֹדֶשׁ*, which could mean *תרומה* (cf. Rashi and Radak), is paralleled with *רִאשִׁית* and *דִּמְעָה*. The term *מְלֶאֱהָ* occurs in two additional places in the Torah:

and it will be considered for you (the Levites) your *תרומה* (i.e., *מַעֲשֵׂר*, the tithes you receive) like the grain from the threshing floor and like the *מְלֶאֱהָ* from the (wine) vat (Num. 18:27).³¹

You shall not sow your vineyard (with) a mixture *הַזֶּרַע הַמְּלֶאֱהָ* (lest

(דִּמְעָה), while in Aramaic with *qān* (i.e., *דִּמְעָה*); cf. *אָמָא*, “cubit,” pl. *אָמוֹת* (H) - *אָמִין* (A).

25 This was already pointed out by Allegro^{pf}.

26 Hoffmann’s edition has *מְלֶאֱהָ*, Epstein’s *מְלִיא*. Since in this work Mekhilta’ refers to the Mekhilta’ deRabbi Ishmael, here is identified so as not to confuse the two Mekhilta’s.

27 The Horovitz - Rabin edition has *מְלֶאֱהָ*, but the variant readings are noted (p. 318-19).

28 The Mishnah assumes that *מְלֶאֱהָ* = *בְּכוֹרִים* and *דִּמְעָה* = *תרומה*; e.g., m. Ter. 3:5.

29 This is rendered in Diez Macho’s edition as, “your tithe and your offering,” and noted on “offering,” “*Dema*’ or *terumah*” (*Neophyti 1* [1970] 2, 147, 474 and n. 5).

30 See A. Tal, *HatTargum hašŠomroni laTorah* (1980) 1, 316-17.

31 In this verse *תרומה* = *מַעֲשֵׂר*, “tithe,” and it is paralleled with *מְלֶאֱהָ*.

you will make unclean the *mele'ah* of the seed) which you planted and the produce of the vineyard" (Deut. 22:9).

The Targumim translate *הָרָעָה הַמֵּלֵאָה* as *פֶּן תִּקְדַּשׁ הָרָעָה* as "that you shall not have to destroy the *mele'ah* of the seed" (TN); *דְּלִמָּא*, "lest you will make (or it will become) unclean the *dema'* of the seed" (TO); and *דְּלִמָּא תַתְחַיֵּיב יִקְדָּתָא דִּימַעַת וְרָעָא*, "lest you shall be obligated to burn the *dema'* of the seed" (PsJ).³²

מִלָּאָה (fem.; *מִלָּא* in JA) and *מָלָא* (masc.) mean "fully ripe produce" or "fullness offering" (from the root *מלא*, "to fill"; cf. *שְׁנֵי דוּדֵינִי דְּוִדִּין מִלָּאִין*, "two pots full of silver," see #17, 4:8). Cf. *אֲכָלוּ כֶּקֶשׁ יֵבֶשׁ מִלָּא*, "they will be consumed like fully ripe dry straw stubble" (Nah. 1:10).³³ Also,

The *בְּכוֹרִים* (first fruit offerings), which are called by four names: *רֵאשִׁית*, *תְּרוּמָה*, *מִלָּאָה*, *בְּכוֹרִים*, have priority over *תְּרוּמָה* (priestly share), which is called only by three names: *רֵאשִׁית*, *תְּרוּמָה*, *דָּמַע*. And *תְּרוּמָה* (priestly share) has a priority over *מַעֲשֵׂר רִאשׁוֹן* (first tithe) since it is called only by two names: *מַעֲשֵׂר* (tithe given to the Levite) and it contains *רֵאשִׁית* (which is) *תְּרוּמַת מַעֲשֵׂר* (priestly share of the tithe given by Levite). And *מַעֲשֵׂר רִאשׁוֹן* (first tithe) has a priority over *מַעֲשֵׂר שֵׁנִי* (second tithe) which is called only by a single name (i.e., *מַעֲשֵׂר*, *Mekhilta'* on Exod. 22:28; cf. m. Ter. 3:7).³⁴

Some commentators equate *דָּמַע* with *תְּרוּמָה*, "priestly share,"³⁵ while others with "tears, dripping," and explain it as "tithe of liquid substances."³⁶ Both

32 *Peruṣ Jonathan* says that *דָּמַע*, the "mixture" of the produce must be burned, and compares it to its Mishnaic meaning. One can reason that planting mixed vegetations eventually will result in burning a produce that contains *terumah* and tithes, thus it is an abomination.

33 Rashi equates *מִלָּא* with *מִלָּאָה* in Exod. 22:28 and Deut. 22:9, and renders in Old-French *אשוּבִיר* *a(s)ovir*, *asobid*, "to be full(y) ripe)" (see Greenberg, 171, 298 n. 517; also Herczeg, *Exodus*, 292 nn. 5-6; Malbim on Exod. 22:28).

34 Zaslansky concludes that *רמֵעַךְ* refers to *מַעֲשֵׂר רִאשׁוֹן*, "first tithe," which is called *דָּמַע* because it is *רְמוּעַ*, "mixed" with *תְּרוּמַת מַעֲשֵׂר*, "priestly share of the tithe," that the Levite must give from the tithe he received from the farmer to the priest, as the Mishnah says, "it contains *רֵאשִׁית*" (m. Ter. 3:7; "מִלֵּאתֶךָ וְרֵמֵעֶךָ" in Sinai 48 [1961], 273-75).

35 E.g., Rashi (on Exod. 22:28; b. Shab. 127b; Rashi offers a different solution, but it is questioned by the Tosaphot in b. Tem. 4a); Rabbi Abraham son of Maimonides (see E.J. Wiesenberg, "Nehor Šeraga" in *Peruṣ Rabbeinu Avraham ben haRambam 'al Berešit uŠemot* [1959], 356-57 and nn. 136-40); Mizrahi says, *תְּבוּאַת הָרָעָה הַנִּקְרָא דָּמַע*, "the yield of the seed which is called *dema'*" (also *Gur 'Aryeh* and *HakKetav wehaqQabbalah*, all on Exod. 22:28); *Ḥatam Sofer* (M. Schreiber - Sofer, *Še'elot uTešuvot Ḥatam Sofer* [1851/52], on 'Even ha'Ezer, Part 2, Tešuvah 170, p. 85); Maharaz (on Num. Rabbah 10:1).

36 E.g., Sa'adiah Ga'on renders *רֵמֵעַ* as *רֵשֶׁח* in Judeo-Arabic, which is translated by Kafah as "the first of the oil"; Kafah also quotes Ibn-Ezra and Rabbi Abraham son of Maimonides (*Peruṣei Rabbeinu Sa'adiah Ga'on 'al haTorah* [1984], 88 and n. 13; see also, M. Zucker, *Rav Saadya Gaon's Translation of the Torah* [in Heb., 1959], 349); Maimonides, Ibn-Saruq in his *Mahberet*, 65, says that *דָּמַע* means pure new wine, Rashbam (refers to Ibn-Saruq; see D. Rosin, *Der Pentateuch-Kommentar des R. Samuel ben Meir*

possibilities are followed by the various interpreters.³⁷ Also,

מְלֹאֲתֶךָ וְדִמְעָךָ (Exod. 22:28) — An additional explanation, מְלֹאֲתֶךָ, these are grapes, because this grape is יָיִן מְלֵאָה (full of wine); וְדִמְעָךָ, these are olives, that do not separate only with difficulty, like the דִּמְעָה (tear) from the eye (Midrash Hafeṣ on Exod. 22:28).³⁸

Most lexicographers derive וְדִמְעָךָ from a secondary meaning of דָּמַע, “to weep,” from which evolved דָּמַע (MH), a synonym for תְּרוּמָה.³⁹ On the other hand, Hoffmann compares דָּמַע of וְדִמְעָךָ to the Samaritan דָּמַע which means חֶלֶב, “fat”; and that דָּמַע is related to the Arabic *dimāḡ*, “brain”; cf. מֶחֶה, “fat,” which is related to מוֹחַ, “brain.”⁴⁰ Kohut connects דָּמַע with the Arabic *dmḡ* which has two meanings: “to mix,” and “to moisten, wet.” The first is reflected in the Mishnaic Hebrew דָּמַע. The biblical וְדִמְעָךָ comes from the second meaning, although the sages applied to it the first meaning.⁴¹ Ben-Yehuda suggests that these are two homophones: דָּמַע, “to shed tears, tears,”

[1881], 119 and n. 6); Rabbeinu Meyuḥas, Sforno, Rosh, Bahya, Malbim, Hirsch (*The Pentateuch* [1960] 2, 382-83).

37 Especially relevant is Kasher’s discussion of the Talmudic and Midrashic sources, as well as the explanations of the commentators (*Torah Shelema* 18, 174-75; on Exod. 22:28). וְדִמְעָךָ is rendered as, “and the outflow of thy/your presses” (JPSⁱ, Cassuto; cf. Pseudo-Jonathan). Cassuto says that דָּמַע is of doubtful etymology, apparently denotes *terumah*, heave-offering, or the priest’s share of the crop (*A Commentary on the Book of Exodus*, 1967). JPSⁱⁱ translates מְלֹאֲתֶךָ וְדִמְעָךָ as “the skimming of the first yield of your vats.” Sarna discusses the various explanations of the agricultural terms מְלֵאָה and דָּמַע, and concludes that their meanings are uncertain. He notes that the Arabic *dimāḡ* is a cognate of דָּמַע. Alternately Sarna offers the translation, “your crop in full bloom, namely, the best part of it.” He refers to Greenfield’s article “The Small Caves of Qumran” (*The JPS Torah Commentary*, Exodus [1991], 140-41, 254b). Koren translates מְלֹאֲתֶךָ וְדִמְעָךָ as, “the first of thy ripe fruits, and of thy liquors”; while *Tanach*, “your fullness-offering or your *terumah*.”

38 Quoted by M. Kasher from a manuscript, *Torah Shelema* 18, 13.

39 MK 1, 300b (mentions the Samaritan דָּמַע for חֶלֶב); BDB, 199b, “(weeping, trickling) juice, i.e., wine or (&) oil”; SB, 174a, 462a (mentions the Samaritan דָּמַע for חֶלֶב); Harkavy, 109a; ESh^{con} 1, 502a; ESh^{dic} 1, 443b. A. Hakham explains that דָּמַע means a cluster of ripe fruits hanging on the tree, separating and falling to the ground. This is like tear drops flowing from the eyes and fall on the ground (*Sefer Šemot Meforaš*, Paršiyot Mišpatim-Pequdei [1991], 56-57). Rieder notes that דָּמַע means ראשית התבואה, “the first of the yield” (*Targum Yonatan ben ‘Uziel ‘al haTorah* [1984] 2, 136).

40 D.Z. Hoffmann, *Mechilta de-Rabbi Simon b. Jochai* (1905), 152 n. 8. As Blau points out, Hoffmann advanced this solution in “Lexicographische Notizen,” *Magazin für die Wissenschaft des Judentums*, 13 (1886), 55. The word מֶחֶה (pl. of the assumed מֶחֶה) occurs twice in the Masoretic Text (Isa. 5:17; Ps. 66:15). In the second case the Targum renders שְׁמָנִין, “fats.” Thus, מֶחֶה means “fats,” and it is related to מוֹחַ, “brain” (see Rashi, Radak, *Mešudat David*, *Mešudat Zion* on Isa. 5:17; Rashi, Ibn-Ezra, *Mešudat Zion* on Ps. 66:15; BDB, 562b; SB, 443a; Harkavy, 365b; BY 4, 2892a and nn. 1-4; ESh^{dic}, 3, 1285a). Some of these comments are mentioned by Greenfield and Sarna (noted earlier).

41 AC 3, 88b-91a.

widely used in Biblical and Mishnaic Hebrew; and דָּמַע, the biblical *hapax legomenon* and the rare Mishnaic Hebrew noun, a synonym of תְּרוּמָה.⁴² Krauss says that דָּמַע is not the same as תְּרוּמָה, rather it is equal to ראשית, which by itself is parallel to קָדֵשׁ, "holy" (cf. Jer. 2:3).⁴³ Ben-Hayyim says that in the Samaritan Hebrew and Aramaic liturgy דָּמַע is used instead of תְּרוּמָה.⁴⁴ Blau suggests that דָּמַע, "tear," is from the root דַּמַּע, where the 'ayin is like the Arabic 'ayn; and דָּמַע, "priestly share," is from the root דַּמַּע, where originally the 'ayin was like the Arabic ḡayn (cf. Arabic "brain"). Both ḡayn and 'ayn merged in the Hebrew 'ayin.⁴⁵ Greenfield agrees with Blau, and separates the two roots. דָּמַע means "the best part, fat part," and applied to cereals. In Mishnaic Hebrew it was used for *terumah*, and not restricted any more to cereals.⁴⁶ Baumgarten agrees with Blau and Greenfield, and says that in 4Q Halaka^a 5:3 the identification of דָּמַע with grain supports the view that originally it meant "the best part, fat part," and was applied to cereals (see below). In Rabbinic sources *dema* was extended to all *terumah* (referring the above Mekhilta's).⁴⁷ Qimron, discussing 4Q Halaka^a, concludes that דָּמַע is ראשית, "first," or תְּרוּמָה, "priestly share," which became specialized for ראשית הַדֶּגֶן, "the first of the grain" (see next paragraph).⁴⁸ Rieder notes that דָּמַע means ראשית הַתְּבוּאָה, "the first of the

42 BY 2, 964-65 n. 5.

43 "Eine neu erschlossene hebräische Vokabel," *Festschrift — Adolph Schwarz zum sibzigsten Geburtstag* (1917), 290-92.

44 Ben-Hayyim points out that Hoffmann compares דָּמַע to the Samaritan דָּמַע. E.g., Eleazar ben Pinehas refers to God as, דָּמַע מְכַל דָּמַע (SH), "Best of the Best"; also דָּמַע דָּמַע דְּרַמְעִין רַכְל אַמְיָה ... דְּבַרְיָאָתָה (SA), "Cream of the creatures ... Creams of all nations" (Marqah 10:13,15), where דְּרַמְעִין, pronounced as *dēmān* (cf. דְּרַמְעִין, "tears," pl. of דָּמַע, "tear," in Aramaic; "פִּיטִים שׁוֹמְרוֹנִים לְשִׁמְחוֹת" in *Tarbiz* 10 [1939], 344 n. 24; *The Literary and Oral Tradition of Hebrew and Aramaic Amongst the Samaritans* [Heb.] 1 & 2, 448 n. 135; 3, part 2, 148 n. 12; 172 n. 15; 194 and n. 15; 315 and n. 13; 336 n. 10).

45 "Über Homonyme und angeblich Homonyme Wurzeln," VT 6 (1956), 246-47. Blau discusses the views of the various scholars. Moreover, the 'ayin in several Hebrew names through Greek is pronounced in the western languages with g. E.g., עֶזְרָה (= 'Azzah), Γαζα, Gaza, Arabic Ḡazze (Gen. 10:19, etc.); עֶמֶרָה (= 'Amorah), Γομωρρα, Gomorrah (Gen. 13:10, etc.). Thus, Hebrew at one time had two types of 'ayins, as Arabic does.

46 Greenfield, 139a.

47 "4Q Halaka^a 5, The Law of *Ḥadash*, and the Pentecontad Calendar," JJS 27 (1976), 36-37.

48 Qimron discusses the various explanations of דָּמַע. He indicates that in the Aramaic text of 4Q529 line 8, דָּמַעָא מִן אֲרָרָא (the dot indicates uncertainty) means דָּגָן מִן הַגֹּרֶן, "grain from the threshing-floor" ("מִלּוֹן הַמִּקְרָא בְּרָאִי הַמְּגִלּוֹת" in *Tarbiz* 58 [1989], 304-07, i). Cf. דָּגָן מִן הַגֹּרֶן is rendered by the Targumim as אֲרָרָא מִן אֲרָרָא (TO, PsJ), "like the grain from the threshing-floor," and אֲרָרָא מִן אֲרָרָא (TN), "like *terumah* of the grain from the threshing-floor" (Num. 18:27; see also Rashi, Rashbam, *HakKetav wehaqQabalalah*). On the other hand, Eisenman and Wise read כְּרִי כְּשֶׁבִין רַמְעָה מִן, and render, "when ... tears from ..." (4Q529 line 8; *Dead Sea Scrolls Uncovered* [1993], 38-39), while Cook translates as "when tears flow freely [...]" (M. Wise, M. Albeg, Jr., E. Cook, *Dead*

yield.”⁴⁹ Sokoloff defines the Aramaic דמע as follows; דמע (masc.), *Teruma*; דמע (verb), “to cry, be tearful”; דמעה (fem.), “tear.”⁵⁰

Milik published a fragment (without translation) of 4Q Halaka^a 5 (now 4Q251 5).⁵¹ The following is an improved interpretation for most of the second and third lines of this text.

הבכורים והמלאה אל יאחר איש כי [תירוש] הוא ראשית המלאה [ו]דגן הוא הדמע
Let no man delay (the offerings of) the *bikkurim* (first fruits) and the *mele'ah*
(fullness of the produce); since (the) [new wine] (offering) is *re'šit hammele'ah*
(the first part of the fullness), [and] (the) grain (offering) is *dema'*.

The second half clarifies the meaning of the first, which paraphrases the above discussed verse מלאאתך ודמעך לא תאחר תירוש (Milik) seems to be accurate; cf. ראשית דגנך תירשך ויצהרך, “the first fruit (i.e., *terumah*) of your grain, new wine, and oil” (Deut. 18:4; see Rashi, Sifri); מעשר דגנך תירשך ויצהרך ובכורת בקרך וצאנך, “the (second) tithe of your grain, your new wine, and your oil and the firstlings of your herd and flock” (Deut. 14:23; see Rashi, Ramban, Sifri; cf. Neh. 10:40, 13:5,12).

The term תרומה (fem.), “elevated gift” (from the root רום, “to lift”) has several meanings, such as “head tax, contribution (to the Tabernacle), priestly share, tithe, dedicated objects, sacred treasure.” Mishnaic Hebrew occasionally uses דמע instead of תרומה. E.g.,

(A priest) who removes earth from a ritually unclean area, may eat from his דמע (priestly share). (A priest) who opens a heap (where a live or dead person is buried) may not eat from his דמע (priestly share) (m. Oh. 16:4).

(A priest) who examines earth of a ritually unclean area (whether there are human bones) may eat his דמע (priestly share) ... (A priest) who examines (the earth) may eat his דמע (priestly share). (But) one who opens a heap (where a live or dead person is buried) may not eat his דמע (priestly share; t. Oh. 16:7-8; cf. y. Ter. 2:4 [9a]).

The Mishnah discusses various levels of טמא ומטמא, ritual “impurity and rendering impurity,” concerning חולין, “unconsecrated food,” תרומה, “priestly share,” and קודש, sacrificial “holy things.” The Mishnah rules that *hullin* contracted with a third level of impurity, הדמע, נאכל בגוד הדמע, “may be consumed in pottage of *dema'*.” *Terumah* contracted with a fourth level of

Sea Scrolls — A new Translation [1966], 427).

49 Targum Yonatan ben 'Uziel 'al haTorah (1984) 2, 136.

50 A Dictionary of Jewish Palestinian Aramaic of the Byzantine Period, 1990, 153a.

51 Milik, 300, “Addenda.” This material have been discussed by Baumgarten (“4Q Halaka^a 5, The Law of *Hadash*, and the Pentecontad Calendar,” *JJS* 27 (1976), 36-46), Qimron (“מילון המקרא בראי המגילות” in *Tarbiz* 58 [1989], 304-07, i), and Eisenman and Wise (*Dead Sea Scrolls Uncovered* [1993], 203, 205).

impurity, נֶאֱכַל בְּנוֹד הַקֹּדֶשׁ, “may be consumed in pottage of *qodeš*.” *Qodeš* contracted with a fifth level of impurity, נֶאֱכַל בְּנוֹד הַקֹּדֶשׁ, “may be consumed in pottage of *qodeš*” (m. Toh. 2:3-5). Thus, *dema*^c (designated as דְּמַע^c) and *terumah* interchangeably, and both are parallel to *qodeš*.⁵²

The verbal forms of דְּמַע^c are common in Mishnaic Hebrew (some are used as nouns): דָּמַע (*Pi^cel*), “makes the mixture like *terumah*,” דָּמַע (*Pu^cal*) and דָּמַע = נִתְּמַע (*Nitpa^cel*), “became a mixture like *terumah*.”⁵³ E.g.,

A *se’ah* (certain measure) of תְּרוּמָה (priestly share) which fell into one hundred (*sa’ahs* of non-sacred produce), and one lifted out (a *se’ah* of the mixture to replace the *terumah*), and (the replaced *terumah*) fell into another place (of non-sacred produce). Rabbi Eliezer says, “מְדַמַּעַת כְּתְרוּמָה וְדֹא” ([The replaced *terumah*] makes the new mixture like real *terumah*).” But the sages say, “(The replaced *terumah*) אֵינָה מְדַמַּעַת (does not make the new mixture like *terumah*) except according to the calculation (of the percentage of real *terumah* in the new mixture). A *se’ah* of *terumah* that fell into less than a hundred (*se’ahs* of non-sacred produce) וְנִדְמַעַו (and [the mixture] became like *terumah*) and it fell from this (mixture which) became like *terumah* into another place (of non-sacred produce). Rabbi Eliezer says, “מְדַמַּעַת כְּתְרוּמָה וְדֹא” ([The replaced *terumah*] makes the new mixture like real *terumah*).” But the sages say, “אֵין הַמְדַּמַּעַת מְדַמַּעַת (A mixture] which became like *terumah* does not make [the new mixture] like *terumah*) except according to the calculation (of the percentage of the real *terumah* in the new mixture)” (m. Ter. 5:5-6; cf. Yalqut Šim‘oni 111).

The Talmud enumerates three states of *terumah*, “priestly share,” or “heave offering.”

- (1) Ritually pure *terumah* — It must be kept in a pure state, protect it from impurities, and present it to a priest. The priests and their households were obligated to consume the *terumah* in ritual purity.
- (2) Ritually impure *terumah* — Although it must be burned, the priests may utilize the generated flame (e.g., oil in a lamp) or fire (e.g., for cooking or heating).
- (3) Questionable *terumah* — Its status as pure or impure is unknown. Such *terumah* can neither be allowed to consume, nor can it be burned. One must put aside questionable *terumah* in a safe place protected from impurities and from animals until it becomes totally useless.

52 Regarding דְּמַע, the commentators refer to Exod. 22:28, where it means תְּרוּמָה (Maimonides, Rash, Bertinoro, *Tiferet Israel*, Kehati; see also Rosh, *Tosephot Yom-Tov*, *Mele’khet Šelomoh*, Ga’on of Vilna, *Mišnah ‘Aharonah*).

53 The denominative verbal formations of תְּרוּמָה in Mishnaic Hebrew are: תָּרַם (*Qal*), “to separate *terumah*” (e.g., m. Ter. 1:1), also “to remove the ash from the altar” (m. Yoma 1:8); נָתַרַם (*Nif^cal*), “to give as a *terumah*” (e.g., m. Sheq. 2:1); הִתְרַם (*Hif^cil*), “to separate *terumah*” (e.g., Mekhilta’ deRabbi Simon bar Yoḥai 20). Interestingly, the so-called “light” formations use the secondary root תָּרַם and deal with the actual *terumah*, while the “heavy” formations (i.e., augmented middle root letter) use the root דְּמַע and deal with the mixture that became like *terumah*. Regarding the root תָּרַם, see Maimonides on m. Ter. 1:1.

Consequently, hiding solid or liquid heave offerings in earthen vessels, and recording them by engraving onto expensive copper sheets, would be illogical. Thus, the suggestion of the use of earthen *dema*^c vessels to store priestly share may be excluded. This conclusion is strengthened by the fact that some of the *dema*^c vessels were hidden in ritually impure grave sites (11:3-4, 11:9-11), clearly unfit for storage sites of any kind of priestly share.

Another synonym for תְּרוּמָה is קָרְבָּן (masc., pl. קָרְבָּנוֹת, MH; e.g., m. Meg. 3:5), also vocalized as קָרְבָּן (Neh. 10:35, 13:31) and קָרְבָּן (Ezek. 40:43), “gift, offering, sacrifice” (from the root קָרַב, “to come near”). In the twelve day long dedication of the Altar of the Tabernacle, each prince of the twelve tribes brought offerings on a separate day. They are listed individually, then the total sum is given (Num. Ch. 7). Among the offerings that included animals were:

וְקָרְבָּנוֹ קִצְרֹת כֶּסֶף אֶחָת שְׁלֹשִׁים וּמֵאָה מִשְׁקָלָהּ מִזֶּרֶק אֶחָד כֶּסֶף שִׁבְעִים שֶׁקֶל בְּשֶׁקֶל
הַקֹּדֶשׁ שְׁנֵיהֶם מִלֵּאִים סֵלֶת בְּלוּלָה בְּשֶׁמֶן לְמִנְחָה: כֹּף אֶחָת עֶשְׂרֵה זָהָב מִלֵּאָה קִטְרֶת: ...
And his offering was: one silver bowl, its weight was a hundred and thirty (shekels); one silver sprinkling basin of seventy shekels, in the sacred shekel; both of them full of fine flour mixed with oil for a meal-offering; one gold ladle (weighing) ten (shekels) full of incense; one golden dish weighing ten shekels, full of incense. ... (Num. 7:13-14, 25-26, etc.).⁵⁴

This detailed listing clearly shows that the various vessels were also קָרְבָּנוֹת, “dedicated gifts” to the Tabernacle (see Appendix E).⁵⁵

OTHER REFERENCES REGARDING דָּמַע

There are other synonymous terms for דָּמַע. E.g., Pharaoh, concerning Jacob and his family, said to Josheph, “and I will give you (pl.) אֶת טוֹב אֶרֶץ מִצְרַיִם (the best of the Land Egypt), and you (pl.) will eat אֶת חֵלֶב הָאָרֶץ (the *helev* [fat] of the land)” (Gen. 45:18).⁵⁶ The Targumim render חֵלֶב as: טוֹבָא, “the good” (TO, TN), שְׁמִנּוּנִיתָא, “the fat” (PsJ), תַּרְב (= תָּרַב, “fat” (STA), and דָּמַע (= דָּמַע), *dema*^c (STJ). Thus, דָּמַע in Samaritan Aramaic is a synonym of חֵלֶב, “fat, good, best.” Likewise, the Vatican manuscript of the Samaritan Targum (Cod. Barberini Or 1) renders מִלֵּאָתָךְ וְדָמַעַךְ as מעצרתך ובדך ותרבך

⁵⁴ וְקָרְבָּנוֹ is rendered by the Targumim as: וְקוֹרְבָּנִיהּ, “and his offering” (TO); וְקוֹרְבָּנִיהּ דִּי קָרַב (PsJ), and וְקוֹרְבָּנִיהּ דִּי קָרַב (Frag.), “and his offering that he offered.” Regarding מִלֵּאָה (sg. fem.), cf. שְׁמֵ שְׁנֵי דוֹרִינוֹרִינֵי מִלֵּאִין כֶּסֶף, “there are two pots full of silver” (see #17, 4:8).

⁵⁵ Levine discusses this list and its system of enumeration (“The Descriptive Tabernacle Texts of the Pentateuch,” *JOS* 35 [1965], 314b-318b). In addition, this biblical list is similar to the list of the Copper Scroll.

⁵⁶ Rashi says on this verse, הוּא, כֹּל חֵלֶב לְשׁוֹן מֵיטֵב הוּא, “Any (instance of) *helev* connotes the “best”; elsewhere Rashi says, הַמּוֹבְחָר שְׁבוּ, חֵלְבוֹ — *helbo* (means) the choicest part that it is it” (Lev. 3:9). See Herczeg’s notes on these verses (*Genesis*, 502 n. 3; *Leviticus*, 30 n. 5).

(= *מַעְצֶרְתְּךָ וּבִדְדֶךָ וּתְרִבְךָ*) "your wine-press and your olive-press and your fat"; i.e., *דָּמַע* is *תִּרְב*, the Aramaic equivalent of *חֶלֶב*, "fat."⁵⁷

חֶלֶב also means *תְּרוּמָה*, "priestly share," *מַעֲשֵׂר*, "tithe" given to the Levite, and *תְּרוּמַת מַעֲשֵׂר*, "priestly share of the tithe" which the Levite must give to the priest. E.g., *כָּל חֶלֶב יִצְהָר וְכָל חֶלֶב תִּירוֹשׁ וְדָגָן*, "All the *helev* (best) of oil and all (the) *helev* (best) of the wine and the grain" (Num. 18:12), is explained by the Rabbis as: *זֶה תְּרוּמָה וְתְרוּמַת מַעֲשֵׂר*, "this is priestly share and priestly share of the tithe" (Sifre on this verse); and *תֵּן חֶלֶב לָזֶה*, "Give *helev* (priestly share and tithe) of this and *helev* (priestly share or tithe) of that (produce separately)" (b. Bek. 53b, see Rashi).

Based on the above discussion of *דָּמַע* and many of its synonyms, one can conclude the followings.

- (1) There are two roots for the word *דָּמַע*: (a) *דָּמַע* which in the verbal form means, "to weep," and in nominal forms "tears, tear drops." (b) *דָּמַע*, which in the nominal form means "best, choicest part," and in verbal forms "to mix," some used as nouns meaning "mixture."
- (2) The terms *דָּמַע*, *בִּכּוּרִים*, *חֶלֶב*, *מִלָּאָה*, *מַעֲשֵׂר*, *קוֹדֵשׁ*, *קֶרֶבֶן*, *רֵאשִׁית*, *תְּרוּמָה*, and *טוֹב*, are synonyms.
- (3) These terms were used in both Biblical and Mishnaic Hebrew.
- (4) The Judeo-Aramaic forms of these terms were used in the Targumim and in the various dialects.
- (5) The Aramaic speaking Jews and Samaritans understood these terms.

RELEVANT REFERENCES TO *כְּלִי־דָמַע*

The following Talmudic citations are relevant to the unusual term *כְּלִי־דָמַע* in the Scroll.

בְּדִי יֵין וּבְדִי שֶׁמֶן הַמְדוּמָעוֹת ([regarding ritual purity,] wine pitchers and oil pitchers which are for mixed things, [the common people in Judea] are trustworthy at the time of (wine) pressing and (olive) crushing (m. Hag. 3:4).

בְּדִי יֵין וּבְדִי שֶׁמֶן הַמְדוּמָעוֹת — Rabbi Yose of the School of Rabbi Bun said, "אלו שקודחין בהן את הדָּמַע" (these [are vessels] in which [they] preserve the *dema'*" (y. Hag. 3:4 [18b]).⁵⁸

In Judea ... (regarding ritual purity), seventy days before the (wine) pressing and (olive) crushing ... (the common people) are trustworthy *עַל הַקּוֹדֵשׁ* (for the

⁵⁷ Tal lists the Targum of *חֶלֶב* by various Samaritan manuscripts as *רַמְעָה*, *רַמְעָה*, *חֶלֶב* ("fat"), *טִיֵּאב* ("good"), *שׁוּמִין* ("fat"), and *תִּרְבָּה* ("fat"; *HatTargum haššomroni laTorah* [1980] 1, 196-97, 316-17.

⁵⁸ *קוֹדֵחִין* is explained as "to keep, store" (cf. *Qorban ha'Edah*, *Penei Mošeh*; Maimonides on the Mishnah); "they draw and separate in them the *terumah*" (Lieberman; see next note). There are two roots: *קָרַח*, "to bore, pierce," and *קָרַח*ⁱⁱ, "to kindle" (cf. BY 6, 5741a and n. 1). Perhaps *קוֹרַח* (sg.) means "it (masc.) keeps warm," and *קוֹדֵחִין* (pl.), "they keep warm"; i.e., oil and wine stored in these vessels are protected from the cold.